

A New Linear Distinguisher for Four-Round AES

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Abstract. In SAC’14, Biham and Carmeli presented a novel attack on DES, involving a variation of Partitioning Cryptanalysis. This was further extended in ToSC’18 by Biham and Perle into the Conditional Linear Cryptanalysis in the context of Feistel ciphers. In this work, we formalize this cryptanalytic technique for Substitution-Permutation Networks and derive several properties. A conditional approximation is then used to approximate the $\text{inv} : GF(2^8) \rightarrow GF(2^8) : x \mapsto x^{2^{54}}$ function which forms the only source of non-linearity in the AES. By extending the approximation to encompass the full AES round function, a linear distinguisher for 4-round AES using $2^{125.72}$ known-plaintexts is constructed; the existence of which is often understood to be impossible. We furthermore demonstrate how to recover 32 key bits directly from this distinguisher with no data or time overhead. In addition to suggesting a new approach to advancing the cryptanalysis of the AES, this result moreover demonstrates a caveat in the standard interpretation of the Wide Trail Strategy — the design framework underlying many SPN-based ciphers published in recent years.

Keywords: Conditional Linear Cryptanalysis · AES · Statistical Distinguisher

1 Introduction

In 1992, Matsui and Yamagishi [MY92] presented a novel cryptanalytic method for block ciphers, which was later generalized by Matsui [Mat93] and named Linear Cryptanalysis. When used to analyze a block cipher, this method is concerned with discovering undesirable linear relations between plaintext and ciphertext bits [Dae95, HN11, Mat93]. When such relations are found, one can launch an attack against the cipher, *e.g.*, Matsui’s Algorithm 1 and 2 key-recovery attacks.

This method has become a staple in the cryptanalysis of block ciphers and has influenced their design ever since. This includes the *Rijndael* cipher [DR20], which went on to become the *Advanced Encryption Standard* (AES) [DBN⁺01] in 2000. This cipher attempts to thwart standard linear cryptanalytic attacks by means of the *Wide Trail Strategy* (WTS) [DR01], the framework underlying its design. Following Rijndael’s standardization as AES in 2000, the WTS has seen more widespread use, for example in the block ciphers Fides [BBK⁺13] and LED [GPPR11], and the hash-function Photon [GPP11]. It introduces non-linear behavior in a cipher by means of small non-linear s-boxes and forces any linear trail to activate many of these. By choosing the number of rounds for a cipher such that the correlation contribution of any trail is sufficiently small, the WTS argues that a cipher then becomes resistant to standard linear cryptanalytic attacks. Applying this argument to AES shows that four rounds can withstand these attacks.

1.1 Our Contribution

In this paper, we present a novel 4-round distinguisher for AES based on a conditioned linear approximation. Our first contribution is the observation that the multiplicative

inverse function over the finite field $\mathbb{F}_{2^8}$ admits a linear approximation with correlation 2^{-1} when the input space is restricted to a certain subspace. Observing that this is the only non-linear operation in AES, we extend this observation to provide a linear approximation of an S-box, and to a full round. Finally, we combine multiple instances of this observation to construct a 4-round conditional linear approximation for AES leveraging a partition of the plaintext space into 2^{32} equivalence classes. This approximation achieves a correlation of magnitude 2^{-43} when applied to the unique *substantial class*, while yielding 0 everywhere else. It is demonstrated that the extent to which the correlation accumulates in a single class is negligibly small for a randomly sampled permutation, indicating this pattern to be a “fingerprint” by which 4-round AES can be distinguished.

It is then observed that the plaintext space partition induces a partition on the key space of the same size. In fact, a plaintext class becomes substantial only when a key from the corresponding key space is used. Discovering the substantial class therefore allows one to extract 32 bits of information on the key.

1.2 Organization

Section 2 commences by stating the notation used throughout this work and necessary core concepts. We attempt to generalize conditional cryptanalysis for SPNs and derive several properties in Section 3. By applying this technique to the AES s-box in Section 4, we demonstrate that it is capable of achieving a 2^{-1} correlation using only a quarter of the data, where the correlation magnitude of traditional linear approximations is known to be upper bounded by 2^{-3} . In Section 5, this approximation is then extended to the full AES round function, yielding a 2^{-16} correlation with all sixteen s-boxes active. When combined with a 3-round linear trail, this forms a novel statistical distinguisher for four-round AES. We furthermore demonstrate how in the known-plaintext model this distinguisher can be exploited to extract a minimum of 32 bits of the key using $2^{125.72}$ data. Since this complexity is beyond reach we are unable to verify its precision. Instead, our empirical validation breaks the approximation into smaller parts and verifies each of them separately showcasing that the distinguisher and the key recovery attacks themselves are valid. A discussion of the implications of these attacks for the WTS are then provided in Section 6 together with directions for future work and concluding words.

2 Preliminaries

2.1 Notation

Let $\mathbb{F}_{2^m}$ denote the finite field of 2^m elements. When $m = 1$, we use 0 and 1 to denote the elements in the group underlying $\mathbb{F}_2$ and refer to them as *bits*; elements in $\mathbb{F}_{2^8}$ are referred to as *bytes*. Bit-strings of length m are used to represent elements in $\mathbb{F}_{2^m}$. Hexadecimal notation is used to shorten the notation of these bit strings, *e.g.* we represent the element $10100111_b \in \mathbb{F}_{2^8}$ as $A7_x$. The binary operation $\oplus : \mathbb{F}_{2^m} \times \mathbb{F}_{2^m} \rightarrow \mathbb{F}_{2^m}$ represents addition on this field, which is commonly known as the *XOR* operation.

The n -dimensional vector space over $\mathbb{F}_{2^m}$ is denoted by $\mathbb{F}_{2^m}^n$. For a vector $v \in \mathbb{F}_{2^m}^n$, $v_i \in \mathbb{F}_{2^m}$ is used to indicate its i th component, with $1 \leq i \leq n$. The binary operation $\oplus : \mathbb{F}_{2^m}^n \times \mathbb{F}_{2^m}^n \rightarrow \mathbb{F}_{2^m}^n : a, b \mapsto (a_1 \oplus b_1, \dots, a_n \oplus b_n)$ denotes the addition of vectors. It will be clear from context whether $\oplus$ is used to denote addition on $\mathbb{F}_{2^m}$ or $\mathbb{F}_{2^m}^n$. Given that $\mathbb{F}_{2^n}$ and $\mathbb{F}_2^n$ are isomorphic vector spaces, we will primarily use $\mathbb{F}_2^n$ to denote the space on n -bit vectors; $\mathbb{F}_{2^n}$ is only used when this is more informative.

We call $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ with $n, m \in \mathbb{N}_+$ a *vectorial Boolean function*. When $n = m$ and F is invertible, F is called a *permutation*. In the special case that $m = 1$, the function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is said to be a *Boolean function*. Note that a vectorial Boolean function F

can be viewed as a vector of Boolean functions $(f_1, \dots, f_m)$ acting on the same input, with $f_i : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ for all $1 \leq i \leq m$. A Boolean function is *linear* if $f(x \oplus y) = f(x) \oplus f(y)$ for all $x, y \in \mathbb{F}_2^n$. Analogously, a vectorial Boolean function F is said to be linear when it satisfies the same equality. Note that this implies that the components of a linear vectorial Boolean function are themselves linear Boolean functions.

We use $a^\top x$ as shorthand notation for the inner product $\langle \cdot, \cdot \rangle : \mathbb{F}_2^n \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2 : a, x \mapsto a_1 x_1 \oplus \dots \oplus a_n x_n$. Observe that for fixed a , we can view $a^\top x$ as a Boolean function on x ; we refer to a as the *mask* of x . The parity functions $a^\top x$ with $a \in \mathbb{F}_2^n \setminus \{0\}$ are the only linear Boolean functions. Consequently, any linear vectorial Boolean function F can be decomposed into a vector of functions $(a_1^\top x, \dots, a_m^\top x)$. Note that this implies that $F(x) = Mx$ for all $x \in \mathbb{F}_2^n$, when the i th row in $M \in \mathbb{F}_2^{m \times n}$ is equal to a_i for all $1 \leq i \leq m$. We will use M_F to denote this matrix for F .

The imbalance of a Boolean function is defined as

$$\text{Imb}(f) := \frac{1}{2}(|\{x \in \mathbb{F}_2^n \mid f(x) = 0\}| - |\{x \in \mathbb{F}_2^n \mid f(x) = 1\}|) \quad (1)$$

where $|\cdot|$ denotes the cardinality of a set. A function with imbalance 0 is said to be *balanced* [DR07]. For the parity functions it holds that $\text{Imb}(a^\top x) = \delta_{a,0} \cdot \frac{1}{2} |\mathbb{F}_2^n| = \delta_{a,0} \cdot 2^{n-1}$, with $\delta_{i,j}$ denoting the *Kronecker delta* function, which is defined as

$$\delta_{i,j} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases} \quad (2)$$

2.2 Linear Cryptanalysis

When used to analyse an n -bit block cipher $E_k(x) = E(x, k) : \mathbb{F}_2^n \times \mathbb{F}_2^\kappa \rightarrow \mathbb{F}_2^n$, standard Linear Cryptanalysis is concerned with discovering masks $u, v \in \mathbb{F}_2^n$ such that the deviation of the probability $p_k := \mathbb{P}[u^\top x = v^\top E_k(x)]$ from $\frac{1}{2}$ is large for almost all keys $k \in \mathbb{F}_2^\kappa$ [Dae95, HN11, Mat93]. This deviation is denoted using ϵ_k and is referred to as the *bias* of the linear approximation (u, v) . However, in this work, we make use of the concept of *correlation* as presented in [Dae95]. Here, the correlation of (u, v) is defined to be

$$C_{v,u} := \mathbb{P}[u^\top x = v^\top E_k(x)] - \mathbb{P}[u^\top x \neq v^\top E_k(x)] = 2\epsilon_k, \quad (3)$$

where one is consequently interested in discovering (u, v) for which the absolute value, or *magnitude*, of this correlation is large for almost all keys.

A common technique to discover such an approximation is through the construction of a *linear trail* [Bih95, HCN19, Mat93]. For an r -round iterated function $F(x) := f_{r-1} \circ f_{r-2} \dots \circ f_0(x)$, a q -round linear trail is a vector of $q+1$ masks $\tau := (\tau_0, \dots, \tau_q) \in (\mathbb{F}_2^n)^{q+1}$ such that τ_i and τ_{i+1} are the input and output masks of round i , respectively. The correlation of the trail is

$$C(\tau) = \prod_{i=0}^q C_{\tau_{i+1}, \tau_i}. \quad (4)$$

In practical setting, the adversary does not have access to intermediate values of the iterated function but only to the input and the output. This gives rise to the notion of a linear hull defined to be the collection of all linear trails with the same input and output masks. In [Nyb95] Nyberg shows that the correlation of a linear hull $\tau_{v,u}$ is the sum of all linear trails with $\tau_0 = u$ and $\tau_{q+1} = v$ over all possible masks for the intermediate values, i.e.,

$$C(\tau_{v,u}) = \sum_{\tau_1, \dots, \tau_q \in \mathbb{F}_2^n} C(\tau). \quad (5)$$

Despite known limitations, it is common in the literature on linear attacks to use the correlation of a single linear trail as a proxy for the correlation of its hull.

2.3 Related Works

2.3.1 Multidimensional Linear Cryptanalysis

Linear Cryptanalysis has been a field of close study, which has led to the inception of a large number of extensions. One particular line of research in this field is concerned with combining multiple approximations and investigating their joint distribution.

The first work in this line is due to Biryukov, De Cannière, and Quisquater in [BDCQ04], which illustrated how to combine statistically independent linear approximations to improve the attack complexity over standard linear cryptanalysis. More recently, Hermelin, Cho and Nyberg introduced *Multidimensional Linear Cryptanalysis* [HCN09, HCN19], where the attacker exploits all linear approximations with linear masks $(u, v) \neq (0, 0)$ in a linear space, thus dropping the independence requirement.

One downside of multidimensional linear cryptanalysis is that requiring the use of an entire linear space restricts the adversary’s ability to choose the best masks for the task at hand. Consequently, the adversary is often forced to include unwanted masks. Such a scenario occurs, for example, when the linear space contains mask pairs with low correlation. Another potential scenario is a result of including linearly dependent masks. Viewed individually, each pair of masks may admit a high correlation, but once the information from enough of these has been extracted, the rest will exhibit diminishing returns.

A third type of unwanted approximations the adversary must consider are those whose input or output mask is zero. Such approximations are clearly needed for analyzing a set of approximations as a linear space, yet they have a negative effect on the attack itself. To avoid this effect, Nyberg further extended multidimensional linear cryptanalysis by designing the *Affine Linear Cryptanalysis* [Nyb19], which offers the option of discarding a whole half-space of linear approximations that does not contribute to statistical non-randomness of the multidimensional linear cryptanalysis.

2.3.2 Conditional Linear Cryptanalysis

In [BP18], Biham and Perle introduced the *Conditional Linear Cryptanalysis*, an extension of Linear Cryptanalysis that conditions linear approximations on other linear approximations. These conditions can be defined by any observable information available: plaintext bits, ciphertext bits, partially encrypted or decrypted data, and any formulae on them. In fact, [BP18] built on a more restrictive variant introduced earlier by Biham and Carmeli in [BC14], where conditioning only involves a plaintext partitioning method that can amplify the correlation of an appropriate linear approximation.

2.3.3 Relations to Previous Work

Our attack is concerned with partitioning the plaintext space into disjoint sets and applying the same linear approximation to each of them separately as suggested in [BC14]. Strictly speaking, this partitioning can be seen as applying a linear approximation with zero as the output mask.¹ Whereas multidimensional linear adversary would treat the information extracted from these approximations the same as it would for any of the other masks, and an affine adversary would ignore this information completely; we use it in a reinforcing way to improve the correlation of another mask.

A second reason why our distinguisher cannot be easily captured by multi-linear models is that having excluded linearly dependent masks we do not form a linear (resp., affine) space at all. We do so after recognizing that most of the useful information is already captured by the approximations we do consider, thus the exclusion of the rest avoids

¹We note however that this is not the only possible framing for this partitioning, nor it is necessarily the most natural one.

facing diminishing returns. At the same time, the masks are not necessarily statistically independent, thus the attack does not fit into the model suggested in [BDCQ04].

To see this difference between our distinguisher and multidimensional ones, consider that partitioning the input space is independent of the key and can even be done in an offline phase (i.e., before the key is even sampled). Alternatively, an adversary may choose to query the oracle for only a single plaintext equivalence class and obtain a 2^{96} -data distinguisher with a success probability of 2^{-32} . An analogous approach is not available for a multidimensional adversary which must query the oracle on uniformly sampled plaintexts and must treat all approximations the same way.

2.4 AES

The Advanced Encryption Standard (AES) [DBN⁺01] is the standardized version of the Rijndael cipher [DR20] that is best described as an r -round key-alternating substitution-permutation network. Its encryption function can be expressed as the composition

$$\text{AES}_k^r := A_{k_r} \circ \text{RF}' \circ A_{k_{r-1}} \circ \text{RF} \circ \dots \circ \text{RF} \circ A_{k_1} \circ \text{RF} \circ A_{k_0}, \quad (6)$$

where $A_{k_i} : \mathbb{F}_2^{128} \rightarrow \mathbb{F}_2^{128} : x \mapsto x \oplus k_i$ denotes the addition of round key k_i and $\text{RF}, \text{RF}' : \mathbb{F}_2^{128} \rightarrow \mathbb{F}_2^{128}$ denote the round functions. In the first $r - 1$ rounds, the round function $\text{RF} := \text{MC} \circ \text{SR} \circ \text{SB}$ is used, while the last round applies $\text{RF}' := \text{SR} \circ \text{SB}$, with $\text{SB}, \text{SR}, \text{MC} : \mathbb{F}_2^{128} \rightarrow \mathbb{F}_2^{128}$.

The standardized cipher specifies three AES variants — AES-128, AES-192, and AES-256 — differing in two aspects. First, the variants involve 10, 12, and 14 rounds respectively. Second, the lengths of the master key k used are 128, 192, and 256 bits. AES includes a specification for the key expansion algorithm used to generate the individual round keys k_i from k . In Section 5 we will use the fact that this algorithm reuses part of this master key as the initial round key k_0 . Aside from this, it is assumed that the round keys generated by this algorithm are statistically independent [LMM91].

The AES state is most conveniently represented using a 4×4 byte matrix, an example of which is provided in Figure 1. This example demonstrates that the 128-bit state may additionally be viewed as a byte-vector $(p_0, \dots, p_{15}) \in \mathbb{F}_{2^8}^{16}$ storing the byte entries of the matrix in column-major order.

p_0	p_4	p_8	p_{12}
p_1	p_5	p_9	p_{13}
p_2	p_6	p_{10}	p_{14}
p_3	p_7	p_{11}	p_{15}

Figure 1: A convenient representation of the AES state.

The vectorial Boolean function **SB**, short for **SubBytes**, applies the s-box $S : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8$ to each of the sixteen bytes. This s-box is best described as the composition $\text{T} \circ \text{L} \circ \text{inv}$. Here, $\text{inv} : \mathbb{F}_{2^8} \rightarrow \mathbb{F}_{2^8} : x \mapsto x^{254} \bmod 11\text{B}_x$ maps an element in $\mathbb{F}_{2^8} \setminus \{0\}$ to its multiplicative inverse modulo 11B_x and zero to itself. The function $\text{L} : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8$ is a linear permutation — its matrix representation is provided in Equation 7 — while $\text{T} : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8 : x \mapsto x \oplus 63_x$ is

an affine translation.

$$M_L = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (7)$$

The **ShiftRows** (SR) operation is a linear function permuting the bytes in the state. Assuming zero-based indexing, this operation cyclically shifts the bytes in the i th row of the state i steps to the left. An illustration is provided in Figure 2.

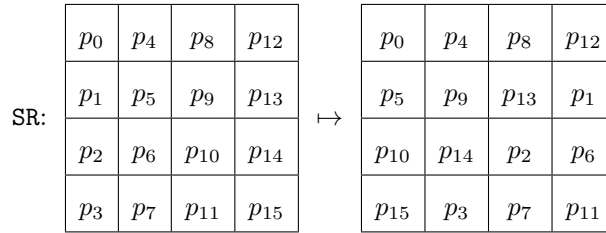


Figure 2: Illustration of the application of **ShiftRows** to the state.

The **MixColumns** (MC) function is a permutation operating on the state column by column. Considering a column as the coefficients of polynomials over $GF(2^8)$, it multiplies each modulo $x^4 + 1$ with the polynomial $c(x) := 3x^3 + x^2 + x + 2$. From the perspective of linear cryptanalysis, it may, however, be more informative to view this operation as a linear transformation $H : \mathbb{F}_2^{32} \rightarrow \mathbb{F}_2^{32}$ that is applied to each of the four 32-bit columns in the state. The matrix representation of H is included as supplementary material.

The **SubBytes** function forms the *substitution part* of this SPN and is the only non-linear component contained in the round function **RF**. As such, one of its purposes is to thwart linear cryptanalysis against the cipher. It is in particular the properties of the non-linear function **inv** that deter linear attacks. Because we will analyze the linear properties of the **inv** function, it may be more insightful to ignore the algebraic structure underlying this function and instead view it as a lookup table. For both **inv** and **S**, these tables are provided as supplementary material.

The **ShiftRows** and **MixColumns** functions form the *permutation* part of this substitution-permutation network. As illustrated in Section 2.5, the purpose of these functions is to force any linear trail through the AES to have many active s-boxes, thus yielding a small contribution to the correlation of the hull containing it.

2.5 The Wide Trail Strategy

The *Wide Trail Strategy* (WTS) was first introduced as a block cipher design philosophy by Daemen in 1995 [Dae95]. Throughout the design of the SHARK [RDP⁺96], SQUARE [DKR97], and BKSQ [DR00] ciphers, the concept was further developed into a broadly-applicable block cipher design framework [DR01], ultimately forming the basis of the Rijndael cipher later standardized as the AES.

In the context of this paper we are concerned with one common (mis)understanding of the wide-trail strategy, i.e., arguing resistance against linear cryptanalysis. To make the argument, a designer analyzes the S-box to find an upper bound on the largest absolute correlation for any pair of linear masks, and likewise analyzes the linear layer (most commonly, an MDS matrix) to find a lower bound on the number of active S-boxes over

any q rounds. Informally, raising the former to the power of the latter results in an upper bound on the highest correlation any q -round linear trail might exhibit. Thus, q can be used to tune the desirable resistance to the attack.

To wit, [DR20] argues that four rounds of AES are sufficient to ensure that any 4-round linear trail for AES has correlation of at most 2^{-75} . Thus, an attack would require at least 2^{150} data, which is not possible for a 128-bit block cipher. Conversely, in the sequel we present a 4-round linear trail which, under very reasonable conditions, admits a correlation of 2^{-43} .

3 Conditional Approximation

In [BC14], Biham and Carmeli show that one can improve the data complexity of a linear attack by partitioning the data into classes and computing the correlation of the approximation for each class individually. The authors illustrate that for properly chosen partitions one will find that the magnitude of the approximation correlation for at least one of the classes is larger than one would observe when computing the correlation using all data. Where Biham and Carmeli only briefly review this technique, we will attempt to formalize it and its notation, and illustrate some of its properties. In this formalization, we restrict ourselves to only conditioning the input of a function; the case for conditioning the output is symmetric and is omitted for brevity. The discussion in this section is made from a mathematical point of view and is purposefully kept abstract. The implications for symmetric-key cryptanalysis and in particular the AES, are deferred to Sections 4–5.

3.1 Definition

A *conditional approximation* $(u, v)|_X$ for a function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ comprises a linear approximation (u, v) with masks $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2^m$, and a vectorial Boolean function $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$. In this approximation, X is leveraged to create a partition $\mathcal{P}_X$ on the set of plaintexts $\mathbb{F}_2^n$ consisting of 2^r classes, with a class $\mathcal{P}_X^b := \{x \in \mathbb{F}_2^n \mid X(x) = b\}$ for each $b \in \mathbb{F}_2^r$. For each class, we are interested in

$$p_b := \mathbb{P}[u^\top x = v^\top F(x) \mid X(x) = b], \quad (8)$$

from which the definition of correlation naturally extends to the definition of *conditional correlation* as

$$C_{v,u}^F \Big|_{\mathcal{P}_X^b} := 2 \cdot \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b] - 1. \quad (9)$$

We furthermore refer to $C^F \Big|_{\mathcal{P}_X^b}$ as a *conditional correlation matrix*. Depending on the choice of X , it may be possible to achieve a conditional correlation in one or more classes that exceeds the correlation of the unconditioned linear approximation. In the following subsections, we provide some general properties of the conditional approximation and relate the conditional approximation to standard linear cryptanalysis. In Section 4, more specific properties will be discussed.

3.2 Properties

In [DGV95, Equation 15] it is demonstrated that for appropriate combinations of vectorial Boolean functions F and G , the correlation matrix $C^{G \circ F}$ can be computed as the matrix product $C^G \cdot C^F$. When using conditional correlation matrices instead, this equality is no longer satisfied. With Lemma 1 and 2 we demonstrate how this equation can be adapted for conditional approximations.

Lemma 1. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^p$, $G : \mathbb{F}_2^p \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For all $b \in \mathbb{F}_2^r$ it holds that*

$$C^{G \circ F} \Big|_{\mathcal{P}_X^b} = C^G \cdot \left(C^F \Big|_{\mathcal{P}_X^b} \right).$$

Proof. The proof of this lemma is deferred to Appendix A.

Lemma 2. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an arbitrary permutation, let $G : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For all $b \in \mathbb{F}_2^r$ it holds that*

$$C^{G \circ F} \Big|_{\mathcal{P}_{X \circ F}^b} = C^G \Big|_{\mathcal{P}_X^b} \cdot C^F.$$

Proof. The proof of this lemma is deferred to Appendix A.

Note here that Lemma 2 only applies to the case that F is a *permutation*.

We furthermore discuss constructing a conditional approximation for the Cartesian product of functions. To this end, we present Lemma 3, where we use $\cdot$ to denote vector concatenation and $\times$ to denote the Cartesian product of vectorial Boolean functions s.t. $(f \times g)(x|y) = f(x)|g(y)$.

Lemma 3. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, $G : \mathbb{F}_2^r \rightarrow \mathbb{F}_2^p$, $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^q$ and $Y : \mathbb{F}_2^r \rightarrow \mathbb{F}_2^s$ be arbitrary vectorial Boolean functions. It holds that*

$$C^F \Big|_{\mathcal{P}_X^a} \otimes C^G \Big|_{\mathcal{P}_Y^b} = C^{F \times G} \Big|_{\mathcal{P}_{X \times Y}^{a|b}}$$

for all $a \in \mathbb{F}_2^q$ and $b \in \mathbb{F}_2^s$, where $\otimes$ denotes the Kronecker product.

Proof. The proof of this lemma is deferred to Appendix A.

Lastly, it was already observed in [BC14] that merging disjoint classes of the same size yields a new class with a conditional correlation that is the average of the original two. We present a generalization of this result in Lemma 4.

Lemma 4. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For any $b, b' \in \mathbb{F}_2^r$ s.t. $b \neq b'$, it holds that*

$$C^F \Big|_{\mathcal{P}_X^b \cup \mathcal{P}_X^{b'}} = \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C^F \Big|_{\mathcal{P}_X^b} + \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C^F \Big|_{\mathcal{P}_X^{b'}}$$

when either $\mathcal{P}_X^b \neq \emptyset$ or $\mathcal{P}_X^{b'} \neq \emptyset$.

Proof. The proof of this lemma is deferred to Appendix A.

Note here in particular that when $|\mathcal{P}_X^b| = |\mathcal{P}_X^{b'}| \neq 0$, the correlation of the merged class is indeed the average of the original two.

3.3 Relation to Standard Linear Cryptanalysis

To provide an understanding of how this version of conditional cryptanalysis fits within the framework of standard linear cryptanalysis, we illustrate a relation between both approximation techniques when applied to Boolean permutations. To this end, we introduce Lemma 5.

Lemma 5. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an arbitrary permutation and $u, v \in \mathbb{F}_2^n$ arbitrary masks. For all $b \in \mathbb{F}_2$ it holds that*

$$(-1)^b \cdot C_{v,0}^F|_{\mathcal{P}_X^b} = C_{v,u}^F,$$

when $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2 : x \mapsto u^\top x$.

Proof. The proof of this lemma is deferred to Appendix A.

It follows from this lemma that a conditional approximation $(0, v)|_{u^\top x}$ restricted to those plaintexts for which $u^\top x = 0$, is equivalent to the linear approximation (u, v) in the sense that both achieve the same correlation when used to approximate the same function. Furthermore, restricting $(0, v)|_{u^\top x}$ to $\mathcal{P}_{u^\top x}^1$ yields a conditional approximation achieving the same correlation as (u, v) , but with opposite sign.

4 Conditional Analysis of the Multiplicative Inverse in $\mathbb{F}_{2^8}$

In this section, we demonstrate a conditional linear approximation of the vectorial Boolean function `inv`. First, we recall in Section 4.1 the seminal work by Nyberg [Nyb94] which provides an upper bound of $2^{1-n/2}$ on the correlation magnitude of any linear approximation for the inverse function on $\mathbb{F}_{2^n}$.

In symmetric-key folklore, these two results are understood to mean that any attempt to approximate this multiplicative inverse in $\mathbb{F}_{2^8}$ must incur a cost of 2^{-3} at the minimum (see *e.g.*, [AAB⁺20, GKM⁺09, MN17, JNPS21]). Conversely, we demonstrate in Section 4.3 how the conditional approximation technique can exhibit a higher correlation when properly applied to `inv`.

4.1 Linear Approximations of the Inversion Function

Nyberg proved in [Nyb94] that the inversion function on finite field $\mathbb{F}_{2^n}$ possesses properties that are highly desirable in cryptography. In particular, it is shown that the correlation of any linear approximation for this function is upper bounded by $2^{1-n/2}$. This property, among others, led Daemen and Rijmen to use it with $n = 8$ as part of the Rijndael s-box [DR20]. Recall that this particular instance of the inversion function was previously introduced in Section 2.4 as `inv` : $\mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8$.

We briefly discuss the linear properties of the `inv` function. To this end, we constructed its correlation matrix C^{inv} . From C^{inv} we observed that the distribution of values in a column $C_{v,\cdot}^{\text{inv}}$ is the same for any output mask $v \in \mathbb{F}_2^8 \setminus \{0\}$. Since `inv` is an involution, we additionally conclude that the same distribution arises when the input mask u is fixed and v varies instead.

In particular, we observe that regardless of the choice of output mask $v \in \mathbb{F}_2^8 \setminus \{0\}$, there always exist exactly five input masks $u \in \mathbb{F}_2^8 \setminus \{0\}$ s.t. $C_{v,u}^{\text{inv}} = 2^{-3}$, where it should be noted that the exact values of these masks depend on the choice of v .² Let us use $\Psi_v = \{\psi_1, \psi_2, \psi_3, \psi_4, \psi_5\}$ to denote the set containing these five input masks. We find that Ψ_v is not linearly independent and has $\text{Rank}(\Psi_v) = 3$. That is, two of the masks can always be expressed in terms of the other three. We moreover observe that one can fix a basis $\Omega_v := \{\omega_1, \omega_2, \omega_3\}$ such that $\{\omega_1, \omega_2, \omega_3, \omega_1 \oplus \omega_2, \omega_1 \oplus \omega_3\} = \Psi_v$. An overview of these bases Ω_v for each $v \in \mathbb{F}_2^8 \setminus \{0\}$ is provided as supplementary material.

²This observation is well known and was first reported, to the best of our knowledge, in [KMT01a, KMT01b].

4.2 Constructing a Conditional Approximation

In [BP18], Biham and Perle briefly discuss the strategies employed to construct the conditional approximations used in their attack for DES. From this discussion, it is evident that the authors primarily make use of heuristic methods and search algorithms, but provide no explicit construction techniques. In this section, we do present a construction method and demonstrate its value in the next subsection, where it is applied to `inv`.

When constructing a conditional approximation for some vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, we propose to select an output mask $v \in \mathbb{F}_2^m$ that has multiple input masks $u_i \in \mathbb{F}_2^n$ for which $|C_{v,u_i}^F|$ is preferably high and at least non-zero, with $1 \leq i \leq r$ and $r \geq 2$. With these u_i and v , we construct the conditional approximation $(0, v)|_X$ with $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r : x \mapsto (u_1^\top x, \dots, u_r^\top x)$.

In this construction, one must keep in mind only to include masks u_i that are linearly independent of one another; when a linearly dependent set is used, some of the classes in $\mathcal{P}_X$ will be empty. We can illustrate this with the example $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^3 : x \mapsto (u_1^\top x, u_2^\top x, (u_1 \oplus u_2)^\top x)$, where the classes $\mathcal{P}_X^{(0,0,1)}$, $\mathcal{P}_X^{(0,1,0)}$, $\mathcal{P}_X^{(1,0,0)}$ and $\mathcal{P}_X^{(1,1,1)}$ are empty due to the linear dependency of the masks. When only linearly independent masks are used, X induces a partition consisting of 2^r classes that all have a size of 2^{n-r} .

We expect this conditional approximation to achieve an absolute correlation on class $\mathcal{P}_X^{\bar{b}}$ that is higher than $|C_{v,u_i}^F|$ for any u_i , when $\bar{b} \in \mathbb{F}_2^r$ s.t. $(-1)^{\bar{b}_i} = \text{sgn}(C_{v,u_i}^F)$ for all $1 \leq i \leq r$. Here, sgn is used to denote the *signum*-function. For the plaintexts x in this class, it namely holds that $u_i^\top x = \bar{b}_i$ and $\mathbb{P}[v^\top F(x) = 0 \mid u_i^\top x = \bar{b}_i] > \frac{1}{2}$ for all $1 \leq i \leq r$.

When the application of this technique yields two classes with the same absolute correlation, it is possible to merge these using Lemma 4 while maintaining the correlation magnitude. To account for the case that the classes one wishes to merge have correlations with opposing signs, we introduce Lemma 6.

Lemma 6. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be an arbitrary vectorial Boolean function, $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2^m$ arbitrary masks, and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r : x \mapsto (u_1^\top x, \dots, u_r^\top x)$ an arbitrary linear vectorial Boolean function with $u_1, \dots, u_r \in \mathbb{F}_2^n$ linearly independent. For all $b \in \mathbb{F}_2^r$ it holds that*

$$(-1)^{b_i} \cdot C_{v,u \oplus u_i}^F \Big|_{\mathcal{P}_X^b} = C_{v,u}^F \Big|_{\mathcal{P}_X^b}.$$

Proof. The proof of this lemma is deferred to Appendix A.

This lemma demonstrates that, with a properly chosen w in the span of the set $\{u_1, \dots, u_r\}$, the signs of the correlations of the two classes one wishes to merge can be made the same when adding w to the input mask of the conditional approximation.

When constructing a conditional approximation for a cipher that starts with a key addition $A_k : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n : x \mapsto x \oplus k$, using a *linear* conditioning function X to partition the plaintext space has another benefit. We can namely use Lemma 2 to see that

$$C^F \Big|_{\mathcal{P}_X^b} \cdot C^{A_k} = C^{F \circ A_k} \Big|_{\mathcal{P}_{X \oplus A_k}^b} = C^{F \circ A_k} \Big|_{\mathcal{P}_{X'}^b}, \quad (10)$$

where $X' : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r : x \mapsto X(x) \oplus X(k)$ constructs the same partition as X : adding k only rearranges the order of the classes such that $\mathcal{P}_X^b = \mathcal{P}_{X'}^{b \oplus X(k)}$. In this situation it thus suffices to construct a conditional approximation for F ; the addition of k only influences *which* class achieves the large correlation.

4.3 Conditional Approximation for `inv`

We now construct a conditional approximation for `inv`. The technique presented in Section 4.2 lends itself exceptionally well to `inv` since we have seen in Section 4.1 that for

any $v \in \mathbb{F}_2^8 \setminus \{0\}$ there are three linearly independent input masks $\omega \in \Omega_v$ s.t. $C_{v,\omega}^{\text{inv}} = 2^{-3}$. Hence, we construct $X : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^3 : x \mapsto (\omega_1^\top x, \omega_2^\top x, \omega_3^\top x)$, which induces a partition with 8 classes that all contain 32 plaintexts. In Table 1 we present an overview of the conditional correlations achieved on each of these eight classes.

Table 1: $C_{v,0}^{\text{inv}}|_{\mathcal{P}_X^b}$ for each $b \in \mathbb{F}_2^3$.

$\omega_1^\top x$	0	0	0	0	1	1	1	1
$\omega_2^\top x$	0	0	1	1	0	0	1	1
$\omega_3^\top x$	0	1	0	1	0	1	0	1
$C_{v,0}^{\text{inv}} _{\mathcal{P}_X^b}$	2^{-1}	2^{-2}	2^{-2}	-2^{-1}	-2^{-3}	-2^{-3}	-2^{-3}	-2^{-3}

From this table we gather that the conditional approximation $(0, v)|_X$ achieves a conditional correlation of $\pm 2^{-1}$ for two plaintext classes. Note that this is a substantial increase when compared to the 2^{-3} correlation upper bound for the linear approximations (ω, v) with $\omega \in \Omega_v$, illustrating the utility of the conditional approximation.

4.4 Improving the Conditional Approximation

Since there are two classes with a correlation of magnitude 2^{-1} , we shall merge these by means of Lemma 4. Before performing this merge, we must first modify the approximation such that the signs of the conditional correlations of these two classes are the same. To this end we leverage Lemma 6 and observe that the correlation sign of the class $\mathcal{P}_X^{(0,1,1)}$ is inverted when we consider the conditional approximation $(\omega_3, v)|_X$ instead of $(0, v)|_X$. The correlations associated with this new conditional approximation are shown in Table 2.

With the signs of the correlations corresponding with $\mathcal{P}_X^{(0,0,0)}$ and $\mathcal{P}_X^{(0,1,1)}$ now the same, it is possible to modify X such that these two classes are merged. To this end, one needs to discover the linear properties that the classes have in common. Observe here that the value of $\omega_1^\top x$ is the same for the two classes of interest and that the value of $\omega_2^\top x \oplus \omega_3^\top x = (\omega_2 \oplus \omega_3)^\top x$ is moreover constant. This means that these two classes are merged when we instead partition on $X' : \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^2 : x \mapsto (\omega_1^\top x, \omega_6^\top x)$, where $\omega_6 := \omega_2 \oplus \omega_3$. The conditional correlations associated with the approximation $(\omega_3, v)|_{X'}$ are presented in Table 3 and shows that the linear approximation (ω_3, v) achieves a correlation of 2^{-1} on plaintext class $\mathcal{P}_{X'}^{\bar{b}}$, with $\bar{b} = (0, 0)$.

It was previously mentioned in Section 4.2 that extending a conditional approximation to include a round-key addition function A_k yields that the conditional correlations are permuted among the classes induced by X , depending on the value of k . This effect is illustrated in Table 4, where the conditional correlations for approximation $(\omega_3, v)|_{X'}$ with $\text{inv} \circ A_k$ are presented. We observe here that the class with the 2^{-1} correlation magnitude is indexed by $\bar{b} \oplus X'(k) = (\omega_1^\top k, \omega_6^\top k)$. It is moreover illustrated that the sign of the correlation is determined by the value of $(-1)^{\omega_3^\top k}$.

Table 2: $C_{v,\omega_3}^{\text{inv}}|_{\mathcal{P}_X^b}$ for each $b \in \mathbb{F}_2^3$.

$\omega_1^\top x$	0	0	0	0	1	1	1	1
$\omega_2^\top x$	0	0	1	1	0	0	1	1
$\omega_3^\top x$	0	1	0	1	0	1	0	1
$C_{v,\omega_3}^{\text{inv}} _{\mathcal{P}_X^b}$	2^{-1}	-2^{-2}	2^{-2}	2^{-1}	-2^{-3}	2^{-3}	-2^{-3}	2^{-3}

Table 3: $C_{v,\omega_3}^{\text{inv}}|_{\mathcal{P}_{X'}^b}$, for each $b \in \mathbb{F}_2^2$.

$\omega_1^\top x$	0	0	1	1
$\omega_6^\top x$	0	1	0	1
$C_{v,\omega_3}^{\text{inv}} _{\mathcal{P}_{X'}^b}$	2^{-1}	0	0	0

Table 4: $C_{v,\omega_3}^{\text{inv} \circ A_k}|_{\mathcal{P}_{X'}^b}$, for $k \in \mathbb{F}_2^8$, $b \in \mathbb{F}_2^2$ and $X' : x \mapsto (\omega_1^\top x, \omega_6^\top x)$.

				$\omega_1^\top x$				
				$\omega_6^\top x$				
	$\omega_3^\top k$	$\omega_1^\top k$	$\omega_6^\top k$					
0	0	0			2^{-1}	0	0	0
	0	1			0	2^{-1}	0	0
	1	0			0	0	2^{-1}	0
	1	1			0	0	0	2^{-1}
1	0	0			-2^{-1}	0	0	0
	0	1			0	-2^{-1}	0	0
	1	0			0	0	-2^{-1}	0
	1	1			0	0	0	-2^{-1}

5 Application to 4-round AES

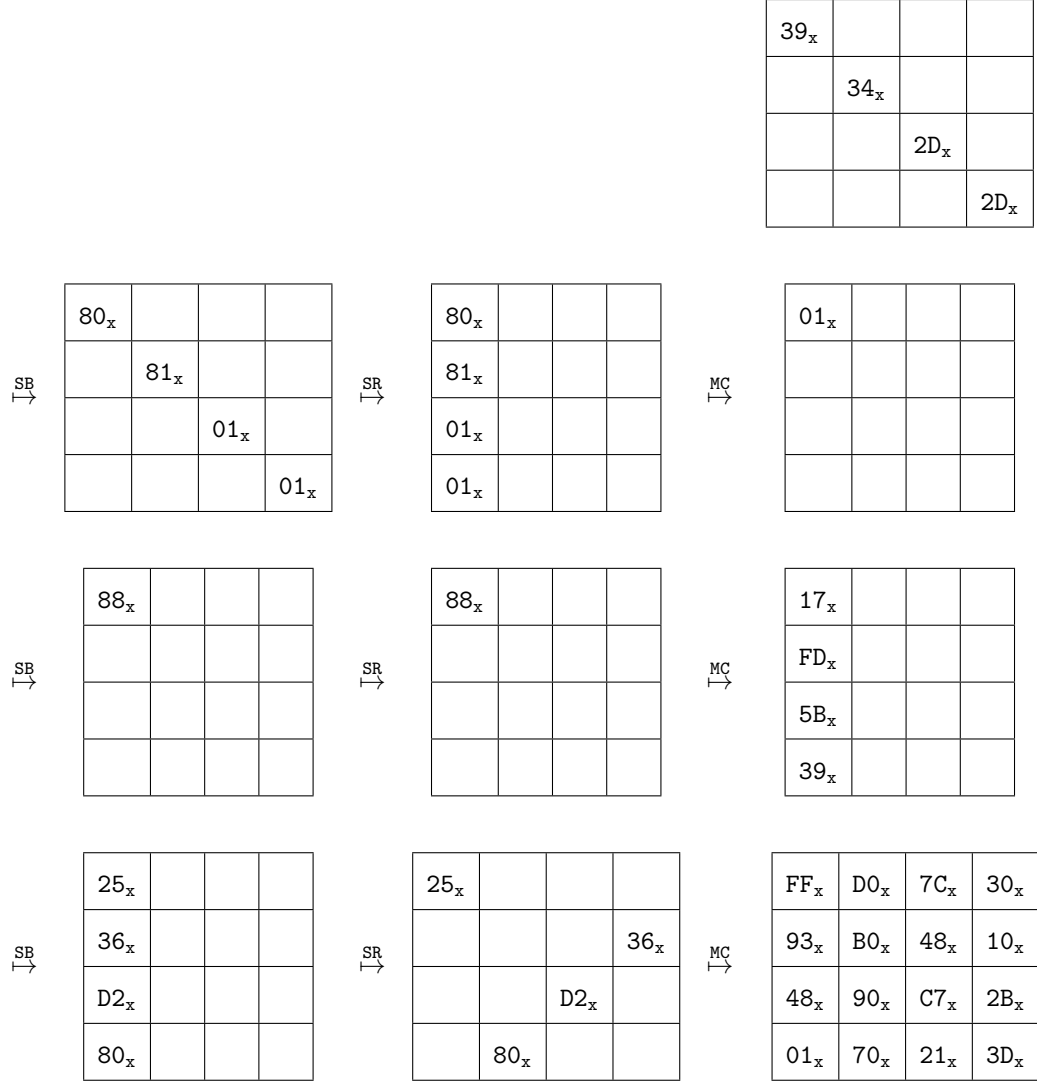
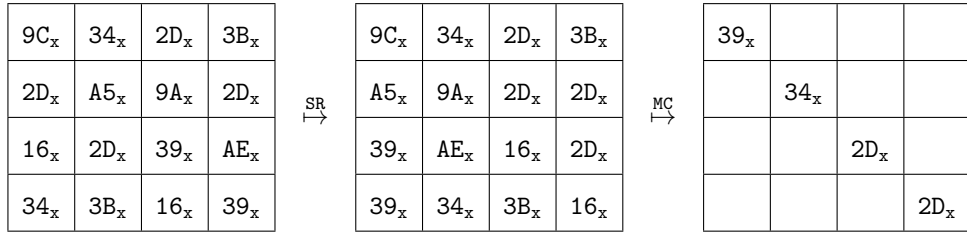
It has been argued that the correlation contribution of a standard linear trail for four-round AES is upper bounded by 2^{-75} [DR20, Section 9.5.2], making it the smallest round-reduced version of this cipher that can withstand distinguishing attacks that rely on standard linear cryptanalysis. However, in symmetric-key folklore this is often incorrectly understood to mean that this round-reduced version of AES can withstand *any form* of linear cryptanalysis. In this section, we present a linear cryptanalysis distinguishing attack against four-round AES, highlighting the need for refining this understanding.

5.1 The four-round distinguisher

A conditional distinguisher for four-round AES is constructed back to front. One commences with constructing a three-round linear trail (v, w) for AES_k^3 . This trail should be formed such that the first and third round have four active s-boxes, while the second round only has one. Moreover, one should make sure that the correlation of the trail over each of the nine s-boxes is 2^{-3} . When done correctly, this yields a trail with correlation $(2^{-3})^9 = 2^{-27}$. In Figure 3, one such example is provided.

Next, one computes the mask u such that the trail (u, v) approximates $\text{MC} \circ \text{SR}$ with correlation 1. Note that u is guaranteed to exist since both MC and SR are linear. In Figure 4 the trail (u, v) is presented.

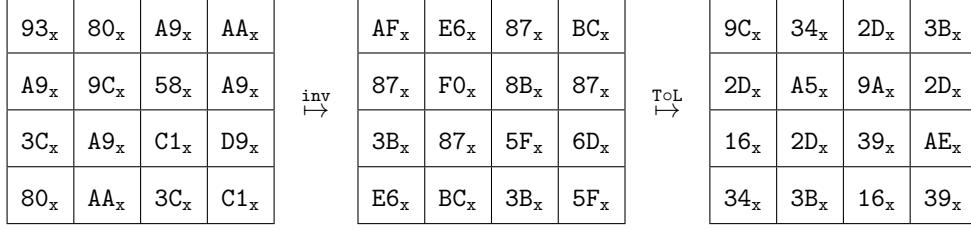
Third, one constructs sixteen trails $([t]_i, [u]_i)$ for sbox component $\text{T} \circ \text{L}$, where we use $[x]_0|[x]_1|\dots|[x]_{15}$ to denote the decomposition of a mask $x \in \mathbb{F}_2^{128}$ into submasks $[x]_i \in \mathbb{F}_2^8$. Note that for each $[u]_i$ there is guaranteed to be a $[t]_i$ such that $([t]_i, [u]_i)$ approximates $\text{T} \circ \text{L}$ with correlation ± 1 since T is affine and L is linear. Following this, one constructs the conditional approximations $([r]_i, [t]_i)|_{X_i}$ with $[r]_i := \omega_3 \in \Omega_{[t]_i}$ and X_i as $x \mapsto (\omega_1^\top x, \omega_6^\top x)$ with $\omega_1, \omega_2, \omega_3 \in \Omega_{[t]_i}$ and $\omega_6 := \omega_2 \oplus \omega_3$. Recalling lemmas 1 and 3, these sixteen trails and sixteen conditional approximation together form the conditional approximation $(r, u)|_X$

Figure 3: Linear trail (v, w) for three-round AES with 9 active s-boxes.Figure 4: Linear trail (u, v) for $\text{MixColumns} \circ \text{ShiftRows}$.

which achieves a conditional correlation with **SubBytes** of magnitude $(2^{-1})^{16} = 2^{-16}$ on plaintext class $\mathcal{P}_X^0$ and 0 on all other classes. Table 5 lists the conditioning masks ω_1 and ω_6 for each of the sixteen conditioning functions X_i , while Figure 5 presents the **SubBytes** trail (r, u) .

Table 5: Masks ω_1 and ω_6 for partitioning functions $X_i : x \mapsto (\omega_1^\top x, \omega_6^\top x)$.

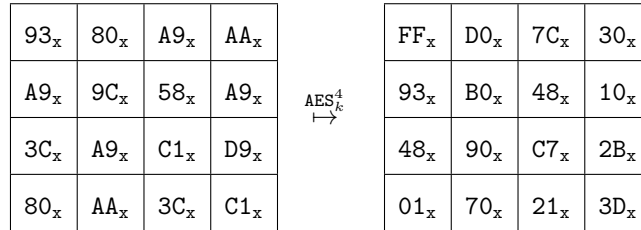
i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
ω_1	E5 _x	93 _x	62 _x	7D _x	7D _x	C0 _x	93 _x	79 _x	93 _x	94 _x	7A _x	62 _x	79 _x	93 _x	9D _x	7A _x
ω_6	3A _x	20 _x	19 _x	EA _x	EA _x	6B _x	20 _x	CE _x	20 _x	49 _x	F9 _x	19 _x	CE _x	20 _x	F2 _x	F9 _x

Figure 5: Linear trail (r, u) for **SubBytes** in first round.

Recalling Lemma 1, one now observes that the conditional approximation $(r, u)|_X$ for **SubBytes** can be concatenated with the trails (u, v) over $\text{MC} \circ \text{SR}$ and (v, w) over AES_k^3 , forming the conditional approximation $(r, w)|_X$. This approximation achieves a conditional correlation of magnitude $2^{-27} \cdot 2^{-16} = 2^{-43}$ with $\text{AES}_k^3 \circ \text{MC} \circ \text{SR} \circ \text{SB}$ on plaintext class $\mathcal{P}_X^0$.

Finally, Equation 10 shows that prepending the round key addition function A_{k_0} yields that $(r, w)|_X$ achieves a conditional correlation with AES_k^4 of magnitude 2^{-43} on plaintext class $\mathcal{P}_X^{b^*}$, where $b^* := X(k_0)$. Moving forward, we will refer to $\mathcal{P}_X^{b^*}$ as the *substantial* plaintext class.

Completing the example, Figure 6 presents the full four-round trail (r, w) and Equation 11 lists the complete conditioning function X .

Figure 6: Linear trail (r, w) for four-round AES with 25 active s-boxes.

5.2 Distinguishing property

We may thus construct a conditional approximation $(u, v)|_X$ with X linear, which, when applied to four-round AES, achieves a correlation of magnitude 2^{-43} when conditioning on the substantial class, while yielding 0 on the other $2^{32} - 1$ classes. We will argue that

this property is highly unique for a 128-bit permutation and thus allows us to distinguish four-round AES from a random permutation.

First, observe that a conditional approximation using a linear conditioning function is akin to considering the correlation of a linear approximation where a number of bits of the input is fixed. In the case of four-round AES, we have shown a set of 32 input bits for which the linear approximation achieves a correlation magnitude of 2^{-43} for a single class (which we refer to as *the substantial class*) and 0 for all other classes; which class is the substantial one depends on the first round-key. To illustrate that this property is unique to AES, it thus suffices to show that when fixing 32 bits of the input of a random permutation, it is highly unlikely that any linear approximation will achieve a correlation of 2^{-43} when restricted to a certain data class.

When fixing 32-bits of the input of a 128-bit permutation P , we can view this function as a set of vectorial Boolean functions $P_1, \dots, P_{2^{32}} : \mathbb{F}_2^{96} \rightarrow \mathbb{F}_2^{128}$, one for each of the 2^{32} possible fixations of the input bits. When P is a random permutation, we argue that these functions P_i are themselves reasonably approximated as random vectorial Boolean functions. It was shown in [DR07] that the correlation of a linear approximation with any vectorial Boolean function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is distributed normally, according to the normal distribution $\mathcal{N}(0, 2^{-n})$. For the functions P_i , this distribution thus evaluates to $N := \mathcal{N}(0, 2^{-96})$. The probability that an arbitrary P_i yields a correlation with magnitude greater than or equal to 2^{-43} with the constructed trail then follows as

$$\mathbb{P}[|X| \geq 2^{-43}] = 2 \cdot \mathbb{P}[X \geq 2^{-43}] = 2 \cdot \mathbb{P}[2^{48} \cdot X \geq 2^5] = 2 \cdot \Phi(-2^5) < 2^{-743.987},$$

where $X \sim N$ and Φ denotes the cumulative density function for the standard normal distribution. Given that the arbitrary permutation P is composed of the 2^{32} functions P_i , the probability that at least one of these P_i achieves the desired correlation can be closely approximated as $2^{-743} \cdot 2^{32} = 2^{-711}$. This probability is small enough to conclude that observing a correlation of 2^{-43} on one of the 2^{32} classes constructed by a linear conditioning function X is a distinguishing property for four-round AES.

5.3 Key-recovery attack

In addition to constructing a distinguishing attack, the conditional approximation $(r, w)|_X$ can be leveraged to construct a key-recovery attack against 4-round AES. We have namely seen that the value of the first round key k_0 fully determines which plaintext class in $\mathcal{P}_X$ becomes substantial. In particular, the substantial class can be indexed by the value of $b^* = X(k_0)$. Since $X : \mathbb{F}_2^{128} \rightarrow \mathbb{F}_2^{32}$, one can thus recover 32 bits of information on this first round key k_0 by determining which class is substantial.

We will demonstrate how the substantial class can be determined in both the chosen and known-plaintext model, as well as derive the data complexity of either attack. Note that these complexity bounds also illustrate the time complexity of both attacks since the data complexity and time complexity are identical in this instance.

5.3.1 Chosen plaintext attack

To determine which plaintext class is substantial, we closely approximate the conditional correlation of all 2^{32} classes and conclude that the greatest absolute correlation corresponds with the substantial class. To construct this approximation in the chosen plaintext model, one selects a sufficient number of plaintext-ciphertext pairs t for each class in the plaintext-ciphertext space partition induced by X , and uses this to compute an empirical approximation of the conditional correlation. To compute the value of t , we introduce a corollary to theorems 2 and 4 in [BN16].

Corollary 1 ([BN16]). *Suppose that the revised wrong-key randomization hypothesis and revised hypothesis of right-key equivalence hold for an n -bit block cipher. Then the*

complexity bound of known-plaintext sampling in Matsui's Algorithm 2 to accomplish an a -bit advantage with a success probability of P_S using a linear approximation achieving a correlation of $C > 2^{-n/2}(\phi_{a+1} + \phi_{P_S})$ is

$$N \geq \frac{(\phi_{P_S} + \phi_{a+1})^2}{C^{-2} - 2^{-n}(\phi_{P_S} + \phi_{a+1})^2} \quad (12)$$

where $\phi_{P_S} := \Phi^{-1}(P_S)$ and $\phi_{a+1} := \Phi^{-1}(1 - 2^{-a-1})$.

Although Blondeau and Nyberg introduce this formula for computing the data complexity of Matsui's Algorithm 2 attack, it can also be applied here: the only difference between these attacks is that ours seeks to find the *plaintext class* achieving the non-zero correlation, whereas Algorithm 2 seeks to find the *class of keys* achieving a non-zero correlation. Since this attack computes correlations with different data sets and a static key, instead of different keys and a static data set, the formula now yields the data complexity per plaintext class. We do note, however, that in our application of the formula, we should choose n as the logarithm of the plaintexts in a class, rather than the bit size of the full cipher.

We determine the number of chosen plaintext-ciphertext pairs necessary to correctly identify the substantial class with 95% certainty. In Table 6, an overview for the data complexity per class d for fixed success probability $P_S = 0.95$ and varying advantages adv is presented. Based on this table, we can conclude that with $2^{92.1}$ plaintext-ciphertext pairs per class one should be able to correctly discern the non-zero correlation plaintext class in at least 95% of experiments. This thus implies a total data complexity of $2^{32} \cdot 2^{92.1} = 2^{124.1}$ chosen plaintext-ciphertext pairs for this attack.

Table 6: Data complexity per plaintext class of a chosen plaintext attack in terms of advantage.

adv	1	8	16	32
$\log_2(d)$	88.44	90.39	91.21	92.09

5.3.2 Known-plaintext attack

The data complexity for the known-plaintext attack is identical, except for the evaluation of the complete data complexity from the complexity per class. In the case of a known-plaintext attack, one has to account for the fact that at least d samples must be encountered for each class even though each sample is uniformly sampled from the full set and therefore corresponds to an arbitrary class.

This is in fact the problem known as the *dixie-cup problem*. Here, one attempts to obtain m copies of n unique objects by uniformly sampling these objects with replacement. It was shown by Newman [New60] that the expected number of objects $E_m(n)$ one must sample before obtaining m instances of each of the n types equals

$$E_m(n) = n \ln n + (m - 1) \cdot n \ln \ln n + \mathcal{O}(1) \quad (13)$$

as n approaches infinity. In our situation, we attempt to encounter $m = d$ samples for $n = 2^{32}$ plaintext classes and are thus interested in the total number of necessary samples $d^* = E_m(n)$. We present the value of d^* in terms of the advantage adv in Table 7. From this table we conclude that $2^{125.72}$ known plaintext-ciphertext pairs will allow one to correctly identify the substantial class with 95% certainty.

Table 7: Expected data complexity in terms of advantage.

adv	1	8	16	32
$\log_2(d^*)$	122.07	124.02	124.84	125.72

5.4 Experimental Validation

We experimentally validate the conditional approximation presented in Section 5.1. Given that the time complexity of the full key recovery attack is $2^{125.72}$, and therefore beyond what we can compute in reasonable time, we instead validate two parts of $(r, w)|_X$ independently.³

5.4.1 Experiment 1

The first experiment restricts itself to the conditional approximation used in the first round of the distinguisher. This approximation with sixteen active s-boxes is expected to achieve a correlation of magnitude 2^{-16} when restricted to data from the substantial class, while achieving a correlation of 0 when restricted to data from other classes.

As such, the experiment generates a random key k and uses this to compute the value of $b^* = X(k_0)$. In addition to this correct class index, fifteen incorrect classes $b_1, \dots, b_{15}$ are randomly selected. For each of the sixteen data classes, $2^{35.62}$ plaintext-ciphertext pairs are generated and used to approximate the conditional correlation for each class. Applying Equation (12) we can deduce that this amount of data would result in the substantial class having the highest correlation among all 16 classes with probability 0.95.

Results The experiment was repeated sixteen times. In fifteen of the sixteen runs, the substantial class was correctly identified. Furthermore, the magnitudes of the correlation observed for the substantial class always fell in the interval $[2^{-16.2620}, 2^{-15.3295}]$ with one outlier at $2^{-17.0648}$. We deem this evidence sufficient to support the claim that the substantial conditional correlation of the conditional approximation $(r, u)|_X$ with AES_k^1 is greater than or equal to 2^{-16} . The details of this experiment and its results are attached as supplementary material.

5.4.2 Experiment 2

To show that a conditional approximation can be followed by (*i.e.*, prepended to) a “standard” linear approximation, we devised a second experiment covering two rounds. We restrict ourselves to a trail $(\bar{r}, \bar{v})|_{\bar{X}}$ which includes eight active S-boxes in the first round and two active S-boxes in the second. The conditional approximation is applied only to the S-boxes in the first round whereas the S-boxes in the second round are approximated in the standard way. The s-boxes have been chosen such that this approximation is expected to achieve a correlation with AES_k^2 of magnitude 2^{-14} when restricted to data from the substantial class, while achieving a correlation of 0 when restricted to data from other classes.

The setup of the experiment is identical to that of Experiment 1, except that the data complexity per class is reduced to $2^{31.62}$.

Results The experiment was executed sixteen times. In fourteen of the sixteen runs, the substantial class was correctly identified. Furthermore, the magnitudes of the correlation observed for the substantial class always fell in the interval $[2^{-15.0900}, 2^{-13.4589}]$. We deem

³A reviewer for this paper noted, and the authors agree, that while this experiment is sufficient for demonstrating the existence of a 4-round linear distinguisher for AES, it is not robust enough to confirm the claimed attack complexities with sufficient precision.

this evidence sufficient to support the claim that the substantial conditional correlation of the conditional approximation $(\bar{r}, \bar{u})|_X$ with AES_k^2 is greater than or equal to 2^{-14} . The details of this experiment and its results are furthermore attached as supplementary material.

6 Discussion and Conclusion

Let us recall that [DR20] argues that the absolute correlation of any 4-round linear trail of AES is upper bounded by 2^{-75} . The widely accepted folklore is that detecting this bias induces a data complexity in the order of 2^{150} which, for a 128-bit block cipher, is not available. Such argumentation has been used extensively to argue the resistance of new SPN-based primitives against linear cryptanalysis.

Conversely, in this paper, we present a statistical distinguisher violating this argument. Concretely, we present a 4-round linear trail for AES which, when applied to a curated subset of plaintexts of size 2^{96} , has correlation $2^{-43} > 2^{-75}$. We then show how to use this trail to construct a linear distinguisher and for recovering 32 key bits all with time and data complexity of $2^{125.72}$.

Building on the work of [KMT01a, KMT01b], we have observed that `inv`, the inversion function on a field with 2^8 elements, has five input masks ω for every output mask $v \in \mathbb{F}_2^8 \setminus \{0\}$ such that $C_{v,\omega}^{\text{inv}} = 2^{-3}$. We then constructed a conditional linear approximation that achieves a conditional correlation of 2^{-1} when applied to `inv`. This approximation was then expanded to encompass a full round of the AES, which was used to construct a statistical distinguisher for four-round AES in the known-plaintext model involving only a single non-trivial linear approximation; the existence of which was understood to be impossible. We have moreover demonstrated a key-recovery attack capable of extracting 32 bits of information on the key using $2^{125.72}$ data. To validate this attack, two small-scale experiments were performed that demonstrated that the conditional approximation can be used to achieve a 4-bit advantage when approximating one-round AES with sixteen active s-boxes and sampling $2^{35.62}$ data for all sixteen plaintext classes, and when approximating two-round AES with 10 active s-boxes while sampling $2^{31.62}$ per class.

With the correctness of the distinguisher established and validated, this work demonstrates that four-round AES is statistically distinguishable from a random permutation in the known-plaintext model using a single non-trivial linear approximation only. In addition to illustrating that this round-reduced version is unsafe, this result presents a potential weakness in any cipher using it as a subroutine (*e.g.*, [MN17, WP13]). Even more impacting is the fact that the vulnerability in four-round AES stems from a caveat in the security argument of the Wide Trail Strategy. This security argument implicitly assumed that, in the case of AES, the correlation magnitude of an approximation for the s-box is upper bounded by 2^{-3} . As such, conditional approximations forms a threat to the security of any of the multitude of ciphers constructed in recent years according to this design framework (*e.g.*, [AAB⁺20, BBI⁺15, BKL⁺07, GNL11, GKR⁺21, JNPS21], *etc.*).

To close, we briefly discuss potential areas of further investigation. For starters, effort should be put towards further improving the correlation achieved by conditional linear cryptanalysis. Moreover, it should be investigated whether augmenting the conditional distinguisher with additional techniques could further improve its performance. Perhaps the distinguisher could be combined with a second trail in a multidimensional manner; the distinguisher might be converted to a zero-correlation distinguisher; or it could be appended to a differential distinguisher after the fourth round. Furthermore, we wonder whether the observations made here can be applied to differential cryptanalysis as well and to other types of s-boxes. Lastly, we point out that this work could potentially be exploited to attack [MN17, WP13] and other cryptographic primitives that use 4-round AES as a subroutine.

Acknowledgements This work is an extension of the second author's Master's thesis [Tak23]. The first author would like to thank Yaniv Carmel who did not contribute anything to this paper but at least hasn't caused harm. The second author would like to thank his parents for their boundless patience; they are still awaiting an invitation to his propaedeutic exam ceremony.

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A Proofs

In this appendix, we present the proofs for all lemmas presented in this work. The proofs are provided in the same order as the lemmas are presented.

Lemma 1. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^p$, $G : \mathbb{F}_2^p \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For all $b \in \mathbb{F}_2^r$ it holds that*

$$C^{G \circ F} \Big|_{\mathcal{P}_X^b} = C^G \cdot \left(C^F \Big|_{\mathcal{P}_X^b} \right).$$

Proof. Let us first observe that for any combination of masks $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2^m$, the correlation of approximation (u, v) with any vectorial Boolean function $H : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ can be computed as

$$C_{v,u}^H = \frac{1}{|\mathbb{F}_2^n|} \sum_{x \in \mathbb{F}_2^n} (-1)^{u^\top x \oplus v^\top H(x)}, \quad (14)$$

while conditional correlation can be computed as

$$C_{v,u}^H \Big|_{\mathcal{P}_X^b} = \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{u^\top x \oplus v^\top H(x)}. \quad (15)$$

We can use these equalities to see that for any $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2^m$, it holds that

$$\begin{aligned} & \sum_{z \in \mathbb{F}_2^p} C_{v,z}^G \cdot C_{z,u}^F \Big|_{\mathcal{P}_X^b} \\ &= \sum_{z \in \mathbb{F}_2^p} \left[\frac{1}{|\mathbb{F}_2^p|} \sum_{x \in \mathbb{F}_2^p} (-1)^{z^\top x \oplus v^\top G(x)} \right] \left[\frac{1}{|\mathcal{P}_X^b|} \sum_{y \in \mathcal{P}_X^b} (-1)^{u^\top y \oplus z^\top F(y)} \right] \\ &= \frac{1}{|\mathbb{F}_2^p|} \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathbb{F}_2^p} \sum_{y \in \mathcal{P}_X^b} (-1)^{u^\top y \oplus v^\top G(x)} \sum_{z \in \mathbb{F}_2^p} (-1)^{z^\top (x \oplus F(y))} \\ &\stackrel{(1)}{=} \frac{1}{|\mathbb{F}_2^p|} \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathbb{F}_2^p} \sum_{y \in \mathcal{P}_X^b} (-1)^{u^\top y \oplus v^\top G(x)} \cdot |\mathbb{F}_2^p| \cdot \delta_{x, F(y)} \\ &= \frac{1}{|\mathcal{P}_X^b|} \sum_{y \in \mathcal{P}_X^b} (-1)^{u^\top y \oplus v^\top G(F(y))} \\ &= C_{v,u}^{G \circ F} \Big|_{\mathcal{P}_X^b}, \end{aligned}$$

where step (1) follows from the fact that the parity function $z^\top \cdot$ is balanced. $\square$

Lemma 2. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an arbitrary permutation, let $G : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For all $b \in \mathbb{F}_2^r$ it holds that*

$$C^{G \circ F} \Big|_{\mathcal{P}_{X \circ F}^b} = C^G \Big|_{\mathcal{P}_X^b} \cdot C^F.$$

Proof. We start with the observation that for any $x \in \mathbb{F}_2^n$ it holds that

$$\begin{aligned} & F(x) \in \mathcal{P}_X^b \\ & \iff F(x) \in \{z \mid X(z) = b\} \\ & \iff x \in \{F^{-1}(z) \mid X(z) = b\} \\ & \iff x \in \{y \mid X(F(y)) = b\} \\ & \iff x \in \{y \mid y \in \mathcal{P}_{X \circ F}^b\} \\ & \iff x \in \mathcal{P}_{X \circ F}^b. \end{aligned} \quad (16)$$

Let us furthermore recall Equation 14 and 15 for computing correlation and conditional correlation, listed in the proof of Lemma 1. We can use these equalities to see that

$$\begin{aligned}
& \sum_{z \in \mathbb{F}_2^n} C_{v,z}^G \Big|_{\mathcal{P}_X^b} \cdot C_{z,u}^F \\
&= \sum_{z \in \mathbb{F}_2^n} \left[\frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{z^\top x \oplus v^\top G(x)} \right] \left[\frac{1}{|\mathbb{F}_2^n|} \sum_{y \in \mathbb{F}_2^n} (-1)^{u^\top y \oplus z^\top F(y)} \right] \\
&= \frac{1}{|\mathcal{P}_X^b|} \frac{1}{|\mathbb{F}_2^n|} \sum_{x \in \mathcal{P}_X^b} \sum_{y \in \mathbb{F}_2^n} (-1)^{u^\top y \oplus v^\top G(x)} \sum_{z \in \mathbb{F}_2^n} (-1)^{z^\top (x \oplus F(y))} \\
&\stackrel{(1)}{=} \frac{1}{|\mathcal{P}_X^b|} \frac{1}{|\mathbb{F}_2^n|} \sum_{x \in \mathcal{P}_X^b} \sum_{y \in \mathbb{F}_2^n} (-1)^{u^\top y \oplus v^\top G(x)} \cdot |\mathbb{F}_2^n| \cdot \delta_{x, F(y)} \\
&= \frac{1}{|\mathcal{P}_X^b|} \frac{1}{|\mathbb{F}_2^n|} \sum_{x \in \mathcal{P}_X^b} \sum_{y \in \mathbb{F}_2^n} (-1)^{u^\top y \oplus v^\top G(x)} \cdot |\mathbb{F}_2^n| \cdot \delta_{F^{-1}(x), y} \\
&\stackrel{(2)}{=} \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{u^\top F^{-1}(x) \oplus v^\top G(x)} \\
&= \frac{1}{|\mathcal{P}_X^b|} \sum_{F(y) \in \mathcal{P}_X^b} (-1)^{u^\top y \oplus v^\top G(F(y))} \\
&\stackrel{(3)}{=} \frac{1}{|\mathcal{P}_X^b|} \sum_{z \in \mathcal{P}_{X \circ F}^b} (-1)^{u^\top z \oplus v^\top G(F(z))} \\
&\stackrel{(4)}{=} C_{v,u}^{G \circ F} \Big|_{\mathcal{P}_{X \circ F}^b},
\end{aligned}$$

where step (1) follows from the fact that the parity function $z^\top \cdot$ is balanced, (2) from the fact that F is a permutation, (3) from Equation 16, and (4) from the fact that $|\mathcal{P}_X^b| = |\mathcal{P}_{X \circ F}^b|$. $\square$

Lemma 3. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, $G : \mathbb{F}_2^r \rightarrow \mathbb{F}_2^p$, $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^q$ and $Y : \mathbb{F}_2^r \rightarrow \mathbb{F}_2^s$ be arbitrary vectorial Boolean functions. It holds that*

$$C^F \Big|_{\mathcal{P}_X^a} \otimes C^G \Big|_{\mathcal{P}_Y^b} = C^{F \times G} \Big|_{\mathcal{P}_{X \times Y}^{a|b}}$$

for all $a \in \mathbb{F}_2^q$ and $b \in \mathbb{F}_2^s$, where $\otimes$ denotes the Kronecker product.

Proof. Let us first observe that $|\mathcal{P}_X^a| \cdot |\mathcal{P}_Y^b| = |\mathcal{P}_{X \times Y}^{a|b}|$ and that

$$y|z \in \mathcal{P}_{X \times Y}^{a|b} \iff y \in \mathcal{P}_X^a \text{ and } z \in \mathcal{P}_Y^b$$

for any $y \in \mathbb{F}_2^n$ and $z \in \mathbb{F}_2^r$. Let $v := v_1|v_2$ and $w := w_1|w_2$ be arbitrary. It then holds that

$$\begin{aligned}
C_{v,w}^{F \times G} \Big|_{\mathcal{P}_{X \times Y}^{a|b}} &= \frac{1}{|\mathcal{P}_{X \times Y}^{a|b}|} \sum_{x \in \mathcal{P}_{X \times Y}^{a|b}} (-1)^{w^\top x \oplus v^\top (F \times G)(x)} \\
&= \frac{1}{|\mathcal{P}_X^a|} \cdot \frac{1}{|\mathcal{P}_Y^b|} \sum_{y|z \in \mathcal{P}_{X \times Y}^{a|b}} (-1)^{(w_1|w_2)^\top (y|z) \oplus (v_1|v_2)^\top (F(y)|G(z))} \\
&= \frac{1}{|\mathcal{P}_X^a|} \cdot \frac{1}{|\mathcal{P}_Y^b|} \sum_{y \in \mathcal{P}_X^a} \sum_{z \in \mathcal{P}_Y^b} (-1)^{w_1^\top y \oplus w_2^\top z \oplus v_1^\top F(y) \oplus v_2^\top G(z)} \\
&= \left[\frac{1}{|\mathcal{P}_X^a|} \sum_{y \in \mathcal{P}_X^a} (-1)^{w_1^\top y \oplus v_1^\top F(y)} \right] \cdot \left[\frac{1}{|\mathcal{P}_Y^b|} \sum_{z \in \mathcal{P}_Y^b} (-1)^{w_2^\top z \oplus v_2^\top G(z)} \right] \\
&= C_{v_1, w_1}^F \Big|_{\mathcal{P}_X^a} \cdot C_{v_2, w_2}^G \Big|_{\mathcal{P}_Y^b}
\end{aligned}$$

with $y \in \mathbb{F}_2^n$ and $z \in \mathbb{F}_2^m$. □

Lemma 4. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r$ be arbitrary vectorial Boolean functions. For any $b, b' \in \mathbb{F}_2^r$ s.t. $b \neq b'$, it holds that*

$$C^F \Big|_{\mathcal{P}_X^b \cup \mathcal{P}_X^{b'}} = \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C^F \Big|_{\mathcal{P}_X^b} + \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C^F \Big|_{\mathcal{P}_X^{b'}}$$

when either $\mathcal{P}_X^b \neq \emptyset$ or $\mathcal{P}_X^{b'} \neq \emptyset$.

Proof. Let $s := |\mathcal{P}_X^b|$. Observe that

$$\begin{aligned}
&\mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&= \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b \setminus \mathcal{P}_X^{b'}] \cdot \mathbb{P}[x \in \mathcal{P}_X^b \setminus \mathcal{P}_X^{b'} \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&\quad + \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'} \setminus \mathcal{P}_X^b] \cdot \mathbb{P}[x \in \mathcal{P}_X^{b'} \setminus \mathcal{P}_X^b \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&\quad + \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'} \cap \mathcal{P}_X^b] \cdot \mathbb{P}[x \in \mathcal{P}_X^{b'} \cap \mathcal{P}_X^b \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&\stackrel{(1)}{=} \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b] \cdot \mathbb{P}[x \in \mathcal{P}_X^b \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&\quad + \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'}] \cdot \mathbb{P}[x \in \mathcal{P}_X^{b'} \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'}] \\
&= \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b] \cdot \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \\
&\quad + \mathbb{P}[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'}] \cdot \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|}
\end{aligned}$$

where step (1) follows from the assumption that $b \neq b'$ and $\mathcal{P}_F$ is a partition. This equality

can now be used to show that for any $u, v \in \mathbb{F}_2^n$ it holds that

$$\begin{aligned}
& C_{v,u}^F \Big|_{\mathcal{P}_X^b \cup \mathcal{P}_X^{b'}} \\
&= 2 \cdot \mathbb{P} \left[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b \cup \mathcal{P}_X^{b'} \right] - 1 \\
&= 2 \left(\mathbb{P} \left[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b \right] \cdot \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \right. \\
&\quad \left. + \mathbb{P} \left[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'} \right] \cdot \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \right) - 1 \\
&\stackrel{(2)}{=} \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot (2\mathbb{P} \left[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^b \right] - 1) \\
&\quad + \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot (2\mathbb{P} \left[u^\top x = v^\top F(x) \mid x \in \mathcal{P}_X^{b'} \right] - 1) \\
&= \frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C_{v,u}^F \Big|_{\mathcal{P}_X^b} + \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} \cdot C_{v,u}^F \Big|_{\mathcal{P}_X^{b'}},
\end{aligned}$$

where step (2) involves the fact that $\frac{|\mathcal{P}_X^b|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} + \frac{|\mathcal{P}_X^{b'}|}{|\mathcal{P}_X^b| + |\mathcal{P}_X^{b'}|} = 1$. Since this holds for all u and v , the lemma follows. $\square$

Lemma 5. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an arbitrary permutation and $u, v \in \mathbb{F}_2^n$ arbitrary masks. For all $b \in \mathbb{F}_2$ it holds that*

$$(-1)^b \cdot C_{v,0}^F \Big|_{\mathcal{P}_X^b} = C_{v,u}^F,$$

when $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2 : x \mapsto u^\top x$.

Proof. Note first of all that it follows from Lemma 4 that

$$C_{v,0}^F \Big|_{\mathcal{P}_X^0} + C_{v,0}^F \Big|_{\mathcal{P}_X^1} = 2 \cdot C_{v,0}^F \Big|_{\mathcal{P}_X^0 \cup \mathcal{P}_X^1} = 2 \cdot C_{v,0}^F \stackrel{(1)}{=} 0,$$

where step (1) follows from the fact that F is a permutation. In particular, note that $C_{v,0}^F \Big|_{\mathcal{P}_X^0} = -C_{v,0}^F \Big|_{\mathcal{P}_X^1}$. Note furthermore that $|\mathcal{P}_X^0| = |\mathcal{P}_X^1| = |\mathbb{F}_2^n|/2$ since X is a balanced function. Subsequently,

$$\begin{aligned}
& C_{v,0}^F \Big|_{\mathcal{P}_X^0} - C_{v,0}^F \Big|_{\mathcal{P}_X^1} \\
&= \left[\frac{1}{|\mathcal{P}_X^0|} \sum_{x \in \mathcal{P}_X^0} (-1)^{0^\top x \oplus v^\top F(x)} \right] - \left[\frac{1}{|\mathcal{P}_X^1|} \sum_{x \in \mathcal{P}_X^1} (-1)^{0^\top x \oplus v^\top F(x)} \right] \\
&\stackrel{(2)}{=} \left[\frac{1}{|\mathcal{P}_X^0|} \sum_{x \in \mathcal{P}_X^0} (-1)^{u^\top x \oplus v^\top F(x)} \right] - \left[-\frac{1}{|\mathcal{P}_X^1|} \sum_{x \in \mathcal{P}_X^1} (-1)^{u^\top x \oplus v^\top F(x)} \right] \\
&= \frac{2}{|\mathbb{F}_2^n|} \sum_{x \in \mathbb{F}_2^n} (-1)^{u^\top x \oplus v^\top F(x)} \\
&= 2 \cdot C_{v,u}^F,
\end{aligned}$$

where step (2) multiplies the summations with $(-1)^{u^\top x}$, which equals 1 for all $x \in \mathcal{P}_X^0$ and -1 for $x \in \mathcal{P}_X^1$. Combining these two results then gives that

$$C_{v,u}^F = \frac{1}{2} (C_{v,0}^F \Big|_{\mathcal{P}_X^0} - C_{v,0}^F \Big|_{\mathcal{P}_X^1}) = \frac{1}{2} (C_{v,0}^F \Big|_{\mathcal{P}_X^0} + C_{v,0}^F \Big|_{\mathcal{P}_X^0}) = C_{v,0}^F \Big|_{\mathcal{P}_X^0} \quad (17)$$

and also $C_{v,u}^F = -C_{v,0}^F \Big|_{\mathcal{P}_X^1}$. $\square$

Lemma 6. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be an arbitrary vectorial Boolean function, $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2^m$ arbitrary masks, and $X : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^r : x \mapsto (u_1^\top x, \dots, u_r^\top x)$ an arbitrary linear vectorial Boolean function with $u_1, \dots, u_r \in \mathbb{F}_2^n$ linearly independent. For all $b \in \mathbb{F}_2^r$ it holds that*

$$(-1)^{b_i} \cdot C_{v, u \oplus u_i}^F \Big|_{\mathcal{P}_X^b} = C_{v, u}^F \Big|_{\mathcal{P}_X^b}.$$

Proof. Let recall Equation 15 provided in the proof of Lemma 1. We can use this to show that

$$\begin{aligned} C_{v, u}^F \Big|_{\mathcal{P}_X^b} &= \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{u^\top x \oplus v^\top F(x)} \\ &= \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{u^\top x \oplus v^\top F(x)} \cdot (-1)^{u_i^\top x} \cdot (-1)^{u_i^\top x} \\ &\stackrel{(1)}{=} (-1)^{b_i} \cdot \frac{1}{|\mathcal{P}_X^b|} \sum_{x \in \mathcal{P}_X^b} (-1)^{(u \oplus u_i)^\top x \oplus v^\top F(x)} \\ &= (-1)^{b_i} \cdot C_{v, u \oplus u_i}^F \Big|_{\mathcal{P}_X^b}. \end{aligned}$$

Here, step (1) follows from the observation that $u_i^\top x = b_i$ for all $x \in \mathcal{P}_X^b$ and that $a^\top x \oplus b^\top x = (a \oplus b)^\top x$. $\square$

Supplementary material

[illegible]

C Lookup Tables

Table 8: Look-up table for $\text{inv}(xy)$.

	y															
	· 0	· 1	· 2	· 3	· 4	· 5	· 6	· 7	· 8	· 9	· A	· B	· C	· D	· E	· F
0 ·	00 _x	01 _x	8D _x	F6 _x	CB _x	52 _x	7B _x	D1 _x	E8 _x	4F _x	29 _x	C0 _x	B0 _x	E1 _x	E5 _x	C7 _x
1 ·	74 _x	B4 _x	AA _x	4B _x	99 _x	2B _x	60 _x	5F _x	58 _x	3F _x	FD _x	CC _x	FF _x	40 _x	EE _x	B2 _x
2 ·	3A _x	6E _x	5A _x	F1 _x	55 _x	4D _x	A8 _x	C9 _x	C1 _x	0A _x	98 _x	15 _x	30 _x	44 _x	A2 _x	C2 _x
3 ·	2C _x	45 _x	92 _x	6C _x	F3 _x	39 _x	66 _x	42 _x	F2 _x	35 _x	20 _x	6F _x	77 _x	BB _x	59 _x	19 _x
4 ·	1D _x	FE _x	37 _x	67 _x	2D _x	31 _x	F5 _x	69 _x	A7 _x	64 _x	AB _x	13 _x	54 _x	25 _x	E9 _x	09 _x
5 ·	ED _x	5C _x	05 _x	CA _x	4C _x	24 _x	87 _x	BF _x	18 _x	3E _x	22 _x	F0 _x	51 _x	EC _x	61 _x	17 _x
6 ·	16 _x	5E _x	AF _x	D3 _x	49 _x	A6 _x	36 _x	43 _x	F4 _x	47 _x	91 _x	DF _x	33 _x	93 _x	21 _x	3B _x
7 ·	79 _x	B7 _x	97 _x	85 _x	10 _x	B5 _x	BA _x	3C _x	B6 _x	70 _x	D0 _x	06 _x	A1 _x	FA _x	81 _x	82 _x
8 ·	83 _x	7E _x	7F _x	80 _x	96 _x	73 _x	BE _x	56 _x	9B _x	9E _x	95 _x	D9 _x	F7 _x	02 _x	B9 _x	A4 _x
9 ·	DE _x	6A _x	32 _x	6D _x	D8 _x	8A _x	84 _x	72 _x	2A _x	14 _x	9F _x	88 _x	F9 _x	DC _x	89 _x	9A _x
A ·	FB _x	7C _x	2E _x	C3 _x	8F _x	B8 _x	65 _x	48 _x	26 _x	C8 _x	12 _x	4A _x	CE _x	E7 _x	D2 _x	62 _x
B ·	0C _x	EO _x	1F _x	EF _x	11 _x	75 _x	78 _x	71 _x	A5 _x	8E _x	76 _x	3D _x	BD _x	BC _x	86 _x	57 _x
C ·	0B _x	28 _x	2F _x	A3 _x	DA _x	D4 _x	E4 _x	0F _x	A9 _x	27 _x	53 _x	04 _x	1B _x	FC _x	AC _x	E6 _x
D ·	7A _x	07 _x	AE _x	63 _x	C5 _x	DB _x	E2 _x	EA _x	94 _x	8B _x	C4 _x	D5 _x	9D _x	F8 _x	90 _x	6B _x
E ·	B1 _x	0D _x	D6 _x	EB _x	C6 _x	0E _x	CF _x	AD _x	08 _x	4E _x	D7 _x	E3 _x	5D _x	50 _x	1E _x	B3 _x
F ·	5B _x	23 _x	38 _x	34 _x	68 _x	46 _x	03 _x	8C _x	DD _x	9C _x	7D _x	A0 _x	CD _x	1A _x	41 _x	1C _x

Table 9: Look-up table for the AES s-box function $S(xy)$

	y															
	· 0	· 1	· 2	· 3	· 4	· 5	· 6	· 7	· 8	· 9	· A	· B	· C	· D	· E	· F
0 ·	63 _x	7C _x	77 _x	7B _x	F2 _x	6B _x	6F _x	C5 _x	30 _x	01 _x	67 _x	2B _x	FE _x	D7 _x	AB _x	76 _x
1 ·	CA _x	82 _x	C9 _x	7D _x	FA _x	59 _x	47 _x	F0 _x	AD _x	D4 _x	A2 _x	AF _x	9C _x	A4 _x	72 _x	C0 _x
2 ·	B7 _x	FD _x	93 _x	26 _x	36 _x	3F _x	F7 _x	CC _x	34 _x	A5 _x	E5 _x	F1 _x	71 _x	D8 _x	31 _x	15 _x
3 ·	04 _x	C7 _x	23 _x	C3 _x	18 _x	96 _x	05 _x	9A _x	07 _x	12 _x	80 _x	E2 _x	EB _x	27 _x	B2 _x	75 _x
4 ·	09 _x	83 _x	2C _x	1A _x	1B _x	6E _x	5A _x	A0 _x	52 _x	3B _x	D6 _x	B3 _x	29 _x	E3 _x	2F _x	84 _x
5 ·	53 _x	D1 _x	00 _x	ED _x	20 _x	FC _x	B1 _x	5B _x	6A _x	CB _x	BE _x	39 _x	4A _x	4C _x	58 _x	CF _x
6 ·	D0 _x	EF _x	AA _x	FB _x	43 _x	4D _x	33 _x	85 _x	45 _x	F9 _x	02 _x	7F _x	50 _x	3C _x	9F _x	A8 _x
7 ·	51 _x	A3 _x	40 _x	8F _x	92 _x	9D _x	38 _x	F5 _x	BC _x	B6 _x	DA _x	21 _x	10 _x	FF _x	F3 _x	D2 _x
8 ·	CD _x	0C _x	13 _x	EC _x	5F _x	97 _x	44 _x	17 _x	C4 _x	A7 _x	7E _x	3D _x	64 _x	5D _x	19 _x	73 _x
9 ·	60 _x	81 _x	4F _x	DC _x	22 _x	2A _x	90 _x	88 _x	46 _x	EE _x	B8 _x	14 _x	DE _x	5E _x	0B _x	DB _x
A ·	E0 _x	32 _x	3A _x	0A _x	49 _x	06 _x	24 _x	5C _x	C2 _x	D3 _x	AC _x	62 _x	91 _x	95 _x	E4 _x	79 _x
B ·	E7 _x	C8 _x	37 _x	6D _x	8D _x	D5 _x	4E _x	A9 _x	6C _x	56 _x	F4 _x	EA _x	65 _x	7A _x	AE _x	08 _x
C ·	BA _x	78 _x	25 _x	2E _x	1C _x	A6 _x	B4 _x	C6 _x	E8 _x	DD _x	74 _x	1F _x	4B _x	BD _x	8B _x	8A _x
D ·	70 _x	3E _x	B5 _x	66 _x	48 _x	03 _x	F6 _x	0E _x	61 _x	35 _x	57 _x	B9 _x	86 _x	C1 _x	1D _x	9E _x
E ·	E1 _x	F8 _x	98 _x	11 _x	69 _x	D9 _x	8E _x	94 _x	9B _x	1E _x	87 _x	E9 _x	CE _x	55 _x	28 _x	DF _x
F ·	8C _x	A1 _x	89 _x	0D _x	BF _x	E6 _x	42 _x	68 _x	41 _x	99 _x	2D _x	0F _x	B0 _x	54 _x	BB _x	16 _x

D Masks

Table 10: The masks $\omega_1, \omega_2, \omega_3 \in \Omega_v$ for every $v \in \mathbb{F}_2^8 \setminus \{0\}$.

v	ω_1	ω_2	ω_3	v	ω_1	ω_2	ω_3	v	ω_1	ω_2	ω_3
01 _x	84 _x	63 _x	F3 _x	29 _x	CD _x	30 _x	B0 _x	51 _x	E2 _x	C9 _x	9D _x
02 _x	7C _x	A2 _x	D2 _x	2A _x	2B _x	54 _x	D9 _x	52 _x	66 _x	98 _x	D8 _x
03 _x	42 _x	31 _x	F9 _x	2B _x	2A _x	6D _x	51 _x	53 _x	49 _x	0A _x	9E _x
04 _x	3E _x	69 _x	D1 _x	2C _x	DA _x	CD _x	30 _x	54 _x	25 _x	2A _x	7B _x
05 _x	5A _x	CD _x	FD _x	2D _x	D3 _x	F1 _x	64 _x	55 _x	95 _x	AA _x	79 _x
06 _x	D4 _x	9A _x	61 _x	2E _x	65 _x	E1 _x	F3 _x	56 _x	15 _x	23 _x	28 _x
07 _x	A1 _x	18 _x	FC _x	2F _x	F5 _x	71 _x	82 _x	57 _x	18 _x	04 _x	E4 _x
08 _x	80 _x	97 _x	DA _x	30 _x	C2 _x	29 _x	2C _x	58 _x	ED _x	8B _x	98 _x
09 _x	9F _x	AB _x	68 _x	31 _x	AB _x	A8 _x	68 _x	59 _x	B8 _x	50 _x	D6 _x
0A _x	AD _x	53 _x	CB _x	32 _x	3F _x	79 _x	5B _x	5A _x	05 _x	08 _x	CA _x
0B _x	4C _x	89 _x	93 _x	33 _x	6F _x	10 _x	E2 _x	5B _x	E9 _x	32 _x	78 _x
0C _x	AE _x	A2 _x	A0 _x	34 _x	8A _x	09 _x	C1 _x	5C _x	B2 _x	F0 _x	F9 _x
0D _x	EA _x	CD _x	B0 _x	35 _x	8F _x	43 _x	49 _x	5D _x	86 _x	81 _x	39 _x
0E _x	D0 _x	DC _x	AE _x	36 _x	3A _x	6C _x	89 _x	5E _x	F2 _x	C9 _x	54 _x
0F _x	C9 _x	44 _x	54 _x	37 _x	C8 _x	38 _x	B2 _x	5F _x	7A _x	38 _x	C1 _x
10 _x	47 _x	33 _x	7B _x	38 _x	F4 _x	C3 _x	AB _x	60 _x	E1 _x	14 _x	96 _x
11 _x	40 _x	8B _x	ED _x	39 _x	B9 _x	F8 _x	E4 _x	61 _x	77 _x	71 _x	63 _x
12 _x	4F _x	9A _x	FB _x	3A _x	36 _x	28 _x	B1 _x	62 _x	3B _x	9D _x	D9 _x
13 _x	9E _x	1B _x	43 _x	3B _x	62 _x	25 _x	3C _x	63 _x	D5 _x	01 _x	61 _x
14 _x	9A _x	60 _x	61 _x	3C _x	8D _x	10 _x	3B _x	64 _x	1F _x	32 _x	A3 _x
15 _x	56 _x	A9 _x	B3 _x	3D _x	89 _x	20 _x	56 _x	65 _x	2E _x	9B _x	FA _x
16 _x	A6 _x	E2 _x	C9 _x	3E _x	04 _x	B9 _x	E0 _x	66 _x	52 _x	58 _x	49 _x
17 _x	CA _x	08 _x	2C _x	3F _x	32 _x	55 _x	91 _x	67 _x	B7 _x	88 _x	F1 _x
18 _x	57 _x	50 _x	D1 _x	40 _x	11 _x	9E _x	CC _x	68 _x	4B _x	09 _x	31 _x
19 _x	7F _x	F2 _x	C9 _x	41 _x	50 _x	39 _x	D6 _x	69 _x	45 _x	41 _x	E0 _x
1A _x	1E _x	99 _x	92 _x	42 _x	03 _x	5C _x	68 _x	6A _x	C7 _x	21 _x	E3 _x
1B _x	75 _x	13 _x	D8 _x	43 _x	26 _x	13 _x	53 _x	6B _x	C1 _x	42 _x	B2 _x
1C _x	E8 _x	BF _x	86 _x	44 _x	7B _x	74 _x	6D _x	6C _x	BA _x	A4 _x	8C _x
1D _x	C4 _x	4A _x	78 _x	45 _x	69 _x	E8 _x	86 _x	6D _x	9D _x	2B _x	D9 _x
1E _x	1A _x	20 _x	6C _x	46 _x	78 _x	67 _x	32 _x	6E _x	E4 _x	F8 _x	59 _x
1F _x	64 _x	22 _x	AA _x	47 _x	10 _x	3B _x	E2 _x	6F _x	33 _x	25 _x	48 _x
20 _x	23 _x	3D _x	99 _x	48 _x	D9 _x	B6 _x	A6 _x	70 _x	AC _x	D0 _x	D2 _x
21 _x	4D _x	6A _x	EA _x	49 _x	53 _x	35 _x	ED _x	71 _x	FA _x	D5 _x	61 _x
22 _x	F6 _x	E9 _x	DB _x	4A _x	AA _x	1D _x	EC _x	72 _x	DC _x	72 _x	A0 _x
23 _x	20 _x	76 _x	E5 _x	4B _x	68 _x	34 _x	A8 _x	73 _x	9C _x	03 _x	6B _x
24 _x	A7 _x	6A _x	DA _x	4C _x	0B _x	A4 _x	B1 _x	74 _x	B6 _x	10 _x	F2 _x
25 _x	54 _x	3B _x	D9 _x	4D _x	21 _x	C2 _x	CA _x	75 _x	1B _x	58 _x	94 _x
26 _x	43 _x	85 _x	94 _x	4E _x	E7 _x	E1 _x	90 _x	76 _x	B1 _x	1E _x	92 _x
27 _x	CF _x	C2 _x	21 _x	4F _x	12 _x	90 _x	F5 _x	77 _x	61 _x	60 _x	4E _x
28 _x	C5 _x	3A _x	93 _x	50 _x	41 _x	18 _x	E4 _x	78 _x	46 _x	5B _x	88 _x

v	ω_1	ω_2	ω_3	v	ω_1	ω_2	ω_3	v	ω_1	ω_2	ω_3
79 _x	BC _x	32 _x	E9 _x	A6 _x	16 _x	48 _x	62 _x	D3 _x	2D _x	91 _x	E9 _x
7A _x	5F _x	68 _x	9F _x	A7 _x	24 _x	21 _x	EB _x	D4 _x	06 _x	63 _x	90 _x
7B _x	44 _x	10 _x	2B _x	A8 _x	83 _x	31 _x	C8 _x	D5 _x	63 _x	71 _x	90 _x
7C _x	02 _x	70 _x	DC _x	A9 _x	92 _x	15 _x	3D _x	D6 _x	E0 _x	41 _x	59 _x
7D _x	E6 _x	C2 _x	29 _x	AA _x	4A _x	55 _x	BC _x	D7 _x	CB _x	ED _x	98 _x
7E _x	DE _x	D0 _x	70 _x	AB _x	31 _x	38 _x	C8 _x	D8 _x	DD _x	1B _x	52 _x
7F _x	19 _x	2A _x	48 _x	AC _x	70 _x	A0 _x	DC _x	D9 _x	48 _x	6D _x	2A _x
80 _x	08 _x	CF _x	E6 _x	AD _x	0A _x	11 _x	94 _x	DA _x	2C _x	08 _x	EB _x
81 _x	F8 _x	45 _x	5D _x	AE _x	0C _x	02 _x	72 _x	DB _x	CE _x	22 _x	5B _x
82 _x	B5 _x	9A _x	FA _x	AF _x	E5 _x	A9 _x	93 _x	DC _x	72 _x	7C _x	AC _x
83 _x	A8 _x	9C _x	6B _x	B0 _x	EE _x	E3 _x	29 _x	DD _x	D8 _x	13 _x	40 _x
84 _x	01 _x	2E _x	B4 _x	B1 _x	76 _x	3A _x	B3 _x	DE _x	7E _x	02 _x	A2 _x
85 _x	98 _x	13 _x	BE _x	B2 _x	5C _x	6B _x	A8 _x	DF _x	99 _x	92 _x	A4 _x
86 _x	5D _x	18 _x	41 _x	B3 _x	A4 _x	B1 _x	92 _x	E0 _x	D6 _x	E8 _x	69 _x
87 _x	93 _x	89 _x	A9 _x	B4 _x	96 _x	12 _x	63 _x	E1 _x	60 _x	4E _x	9B _x
88 _x	8E _x	E9 _x	78 _x	B5 _x	82 _x	84 _x	65 _x	E2 _x	51 _x	16 _x	62 _x
89 _x	3D _x	0B _x	BA _x	B6 _x	74 _x	19 _x	48 _x	E3 _x	FD _x	B0 _x	6A _x
8A _x	34 _x	C0 _x	C3 _x	B7 _x	67 _x	4A _x	91 _x	E4 _x	6E _x	39 _x	50 _x
8B _x	94 _x	11 _x	58 _x	B8 _x	59 _x	F8 _x	FC _x	E5 _x	AF _x	BA _x	23 _x
8C _x	FF _x	1A _x	93 _x	B9 _x	39 _x	3E _x	D6 _x	E6 _x	7D _x	6A _x	80 _x
8D _x	3C _x	33 _x	19 _x	BA _x	6C _x	20 _x	89 _x	E7 _x	4E _x	01 _x	FB _x
8E _x	88 _x	1D _x	79 _x	BB _x	C3 _x	C0 _x	9C _x	E8 _x	1C _x	59 _x	E0 _x
8F _x	35 _x	40 _x	D8 _x	BC _x	79 _x	64 _x	AA _x	E9 _x	5B _x	79 _x	88 _x
90 _x	9B _x	D5 _x	D4 _x	BD _x	BF _x	81 _x	D1 _x	EA _x	0D _x	21 _x	CA _x
91 _x	EC _x	D3 _x	5B _x	BE _x	CC _x	49 _x	52 _x	EB _x	97 _x	30 _x	DA _x
92 _x	A9 _x	1A _x	76 _x	BF _x	BD _x	A1 _x	E0 _x	EC _x	91 _x	4A _x	C4 _x
93 _x	87 _x	0B _x	28 _x	C0 _x	F0 _x	4B _x	7A _x	ED _x	58 _x	49 _x	D7 _x
94 _x	8B _x	AD _x	75 _x	C1 _x	6B _x	34 _x	F4 _x	EE _x	B0 _x	97 _x	30 _x
95 _x	55 _x	DB _x	A3 _x	C2 _x	30 _x	27 _x	7D _x	EF _x	FC _x	F8 _x	B9 _x
96 _x	B4 _x	9A _x	60 _x	C3 _x	BB _x	31 _x	38 _x	F0 _x	C0 _x	F7 _x	9C _x
97 _x	EB _x	05 _x	08 _x	C4 _x	1D _x	D3 _x	F1 _x	F1 _x	A3 _x	2D _x	C4 _x
98 _x	85 _x	52 _x	58 _x	C5 _x	28 _x	23 _x	99 _x	F2 _x	5E _x	19 _x	2A _x
99 _x	DF _x	1A _x	20 _x	C6 _x	FE _x	D8 _x	CB _x	F3 _x	FB _x	01 _x	D5 _x
9A _x	14 _x	12 _x	82 _x	C7 _x	6A _x	80 _x	B0 _x	F4 _x	38 _x	B2 _x	C1 _x
9B _x	90 _x	F5 _x	71 _x	C8 _x	37 _x	AB _x	9F _x	F5 _x	2F _x	60 _x	9B _x
9C _x	73 _x	F0 _x	C8 _x	C9 _x	0F _x	19 _x	51 _x	F6 _x	22 _x	88 _x	95 _x
9D _x	6D _x	62 _x	51 _x	CA _x	17 _x	5A _x	EA _x	F7 _x	F9 _x	F0 _x	8A _x
9E _x	13 _x	40 _x	AD _x	CB _x	D7 _x	0A _x	11 _x	F8 _x	81 _x	B8 _x	6E _x
9F _x	09 _x	7A _x	C8 _x	CC _x	BE _x	8B _x	40 _x	F9 _x	F7 _x	03 _x	AB _x
A0 _x	A0 _x	AC _x	D2 _x	CD _x	29 _x	2C _x	24 _x	FA _x	71 _x	14 _x	82 _x
A1 _x	07 _x	B8 _x	D1 _x	CE _x	DB _x	C4 _x	78 _x	FB _x	F3 _x	E1 _x	14 _x
A2 _x	D2 _x	D0 _x	0C _x	CF _x	27 _x	5A _x	80 _x	FC _x	EF _x	E8 _x	B8 _x
A3 _x	F1 _x	CE _x	64 _x	D0 _x	0E _x	70 _x	A2 _x	FD _x	E3 _x	05 _x	29 _x
A4 _x	B3 _x	4C _x	6C _x	D1 _x	A5 _x	A1 _x	18 _x	FE _x	C6 _x	0A _x	52 _x
A5 _x	D1 _x	50 _x	B8 _x	D2 _x	A2 _x	A0 _x	70 _x	FF _x	8C _x	99 _x	28 _x

E Experiment details

This appendix discusses the three experiments in detail and presents the results of each.

E.1 Experiment 1

The first experiment verifies the conditional approximation $(r, u)|_X$ used for the first round. We present the linear trail part (r, u) of this approximation in Figure 7. Note that this trail is formed as the concatenation of the trails presented in figures 5 and 4. The conditioning function X is identical to the one found in Equation 11.

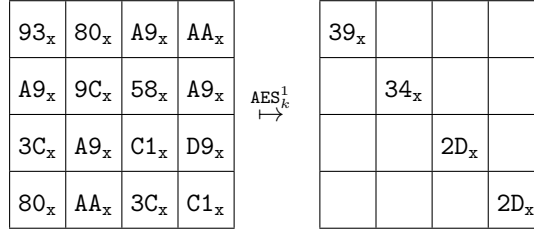


Figure 7: Linear trail (u, v) for one-round AES with 16 active s-boxes.

For this experiment, it was decided to compute an approximation of the correlation of (r, u) with AES_k^1 for sixteen of the 2^{32} data classes induced by X : $X(k_0)$, the class expected to achieve a 2^{-16} conditional correlation, and fifteen arbitrary classes, each expected to achieve a correlation of 0. Leveraging Corollary 12, we find that sampling

$$\begin{aligned} N &= (\Phi^{-1}(P_S) + \Phi^{-1}(1 - 2^{-a-1}))^2 \cdot C^{-2} \\ &= (\Phi^{-1}(0.95) + \Phi^{-1}(1 - 2^{-a-1}))^2 \cdot (2^{-16})^{-2} \\ &= 12.31 \cdot 2^{32} \approx 2^{35.62} \end{aligned}$$

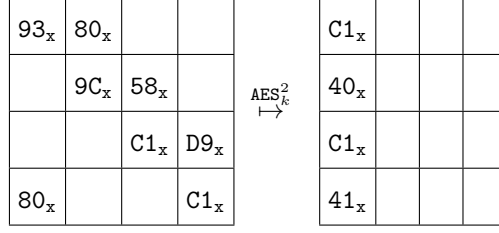
uniformly random sampled plaintext-ciphertext pairs per class, should allow $X(k_0)$ to achieve the greatest correlation magnitude of all sixteen classes with a probability of 0.95 if its correlation is indeed 2^{-16} and 0 for the other, arbitrarily chosen classes. Thus, we

1. sample $2^{35.62}$ data for each class,
2. use this data to approximate the correlation of (u, v) with AES_k^1 for each class, and
3. verify that the correlation corresponding with the correct data class has the greatest magnitude of the sixteen classes.

The experiment was run sixteen times. The randomly generated encryption keys k as well as the IDs of the classes $X(k_0)$ for each run are presented in Table 11. An overview of the observed correlation approximations for each class are found in Tables 12–15.

E.2 Experiment 2

The second experiment verifies the conditional approximation $(\bar{r}, \bar{v})|_{\bar{X}}$ used for the first two rounds. We present the linear trail part $(\bar{r}, \bar{v})$ of this approximation in Figure 8. Note that this trail is formed as an extension of partial the concatenation of the trails presented in figures 5, and 4, and is expected to achieve a correlation of 2^{-14} on the substantial class. The conditioning function X is a modified version of the one found in Equation 11, and is presented in Equation 18.

Figure 8: Linear trail $(\bar{r}, \bar{v})$ for two-round AES with 10 active s-boxes.

$$\begin{aligned}
X : \mathbb{F}_2^{128} &\rightarrow \mathbb{F}_2^{16} : x \mapsto \\
&(\text{E5 } 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 7D \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 7D \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ C0 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 94 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 7A \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 9D \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 7A_x^\top x, \quad (18) \\
&3A \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ EA \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ EA \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 6B \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 49 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ F9 \ 00 \ 00 \ 00 \ 00 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ F2 \ 00_x^\top x, \\
&00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ 00 \ F9_x^\top x)
\end{aligned}$$

For this experiment, it was decided to compute an approximation of the correlation of $(\bar{r}, \bar{v})$ with AES_k^2 for sixteen of the 2^{32} data classes induced by X : $X(k_0)$: the substantial class and fifteen arbitrary classes expected to achieve a correlation of 0. Leveraging Corollary 12, we find that sampling

$$\begin{aligned}
N &= (\Phi^{-1}(P_S) + \Phi^{-1}(1 - 2^{-a-1}))^2 \cdot C^{-2} \\
&= (\Phi^{-1}(0.95) + \Phi^{-1}(1 - 2^{-a-1}))^2 \cdot (2^{-14})^{-2} \\
&= 12.31 \cdot 2^{28} \approx 2^{31.62}
\end{aligned}$$

uniformly random sampled plaintext-ciphertext pairs per class, should allow the substantial class to achieve the greatest correlation magnitude of all sixteen classes with a probability of 0.95 if its correlation is indeed 2^{-14} and 0 for the other, arbitrarily chosen classes.

The experiment was run sixteen times. The randomly generated encryption keys k as well as the IDs of the classes $X(k_0)$ for each run are presented in Table 16. An overview of the observed correlation approximations for each class are found in tables 17 and 18.

Table 11: Settings Experiment 1

run	key k	substantial class $X(k_0)$
1	88 0E E0 4B BF 21 F4 33 13 A3 8D FE 49 59 3B FB _x	D9FB9362 _x
2	8B 7A EB E0 59 3F 1B C7 69 05 9B 19 A6 98 56 2F _x	22E950DB _x
3	D6 B6 42 AF 31 1F C6 E3 0A 11 6A 19 89 02 B3 CE _x	C9C74D71 _x
4	34 73 51 C4 88 FD 52 31 3E A4 BF 25 81 58 70 89 _x	691F40BB _x
5	5C 75 EF 33 F4 83 1C 39 2D 50 8A 7D 23 E7 61 2A _x	36C749DC _x
6	6E 24 F2 80 74 B3 DD B1 AC 1C 69 AE 87 CF 3A E4 _x	A7ACF5DB _x
7	79 F4 39 C6 0E E0 A9 25 A7 2A 55 6F C1 25 63 99 _x	A096F3E6 _x
8	FB 1B 90 FE F3 6A FE E9 7A 14 B0 BA EF EB 51 D3 _x	560533AC _x
9	3C A5 5A 91 15 CF E2 17 36 9C 77 61 F1 94 CF 89 _x	0841D2D3 _x
10	40 B7 FE 84 B3 C2 E6 CB 17 5A 59 DF 6A 9B 62 13 _x	B9E85A1A _x
11	5C E2 A9 5B 40 85 49 F5 49 7D C5 A9 23 17 95 9B _x	2EB55D68 _x
12	F3 A3 74 6B 7C 95 34 2B 08 A4 CB 07 85 78 CA 43 _x	4F3CEE16 _x
13	68 D8 84 4F EB 86 F0 85 28 A0 06 17 F5 11 21 B6 _x	05731E8B _x
14	F8 29 2B 7E A4 06 98 88 B3 2A 02 98 DD B1 88 29 _x	D124C4C7 _x
15	88 C0 B6 C1 34 7B 85 C0 6A E3 B6 3E A2 B4 0C 35 _x	CDE8ACA5 _x
16	5D FA 71 65 47 00 57 D6 8E EC 35 C1 7D 62 73 69 _x	CA1D403C _x

Table 12: Results Experiment 1, run 1-3

Run 1		Run 2		Run 3	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
D9FB9362 _x	$2^{-15.4999}$	22E950DB _x	$2^{-16.1102}$	C9C74D71 _x	$-2^{-15.9625}$
D818A478 _x	$2^{-19.1573}$	CE3417E4 _x	$2^{-18.1270}$	5B05CFDC _x	$-2^{-19.3660}$
E605FB82 _x	$-2^{-17.9155}$	B8CB982E _x	$2^{-19.8481}$	D4F596BC _x	$2^{-17.2500}$
B2674FB6 _x	$-2^{-17.6730}$	04857711 _x	$-2^{-18.5216}$	8E3CEED7 _x	$-2^{-18.0353}$
8A3B6341 _x	$2^{-17.5305}$	0E9736CE _x	$2^{-17.9451}$	2527A236 _x	$2^{-21.0380}$
01AAA9D9 _x	$2^{-19.5722}$	30C1E6DA _x	$2^{-17.7053}$	BEAA017C _x	$2^{-19.7306}$
3E3CCD29 _x	$-2^{-17.7783}$	E715387C _x	$-2^{-17.6672}$	FE483590 _x	$2^{-17.4687}$
4C4E58B7 _x	$-2^{-22.4873}$	F854C41D _x	$-2^{-17.9095}$	8EE41D6D _x	$-2^{-18.2623}$
7DCDAD8B _x	$2^{-19.7293}$	03FA8B0E _x	$2^{-17.8999}$	462E26EE _x	$-2^{-19.2456}$
748D48A9 _x	$2^{-17.3644}$	CFF74618 _x	$2^{-21.4722}$	1C0B28EB _x	$2^{-17.9268}$
BE90D164 _x	$-2^{-17.6962}$	146146C7 _x	$-2^{-18.5509}$	DBAFA19B _x	$2^{-17.4078}$
4527B12D _x	$-2^{-17.5678}$	CA06AB96 _x	$-2^{-22.9214}$	845A6D19 _x	$-2^{-18.1232}$
987ABE46 _x	$-2^{-18.0745}$	A6846144 _x	$2^{-18.6522}$	90E62B87 _x	$-2^{-16.8759}$
83945F76 _x	$2^{-17.3502}$	394B5D83 _x	$-2^{-16.5908}$	A77796EB _x	$-2^{-16.9372}$
E9B0845F _x	$2^{-18.0487}$	1CC311A6 _x	$-2^{-18.5728}$	A62113EF _x	$2^{-17.8797}$
A087006E _x	$-2^{-18.1554}$	F7AA8286 _x	$-2^{-20.4581}$	12BC7FF7 _x	$-2^{-18.6994}$

Table 13: Results Experiment 1, run 4-12

Run 4		Run 5		Run 6	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
691F40BB _x	-2 ^{-15.7756}	36C749DC _x	2 ^{-16.2046}	A7ACF5DB _x	-2 ^{-15.3295}
16E3FDBE _x	2 ^{-23.7022}	FBCFEAC8 _x	2 ^{-18.3684}	3723281B _x	2 ^{-19.9286}
B3897963 _x	2 ^{-18.7780}	61CA4EFA _x	2 ^{-19.1917}	A7791C20 _x	2 ^{-17.6121}
93CFEF88 _x	2 ^{-17.3957}	66F33BE1 _x	2 ^{-19.5200}	57D0D38A _x	-2 ^{-20.2062}
A7B89A0C _x	-2 ^{-16.8270}	E4C79FDF _x	2 ^{-18.5953}	2E8AA8D0 _x	-2 ^{-17.9525}
FB5C4923 _x	-2 ^{-17.1264}	51B30397 _x	-2 ^{-20.0508}	AF73613D _x	-2 ^{-18.0425}
4B257145 _x	2 ^{-18.7155}	6E6105C0 _x	2 ^{-17.9348}	2127A29C _x	2 ^{-18.3473}
59C4690F _x	-2 ^{-18.0092}	CC14343D _x	-2 ^{-18.8991}	B999CDA8 _x	2 ^{-18.4215}
E75E61C7 _x	2 ^{-18.1044}	89A6C359 _x	2 ^{-19.5868}	9B722706 _x	2 ^{-17.0399}
7108E4AF _x	2 ^{-19.0402}	4921655D _x	2 ^{-17.4314}	ABB3564F _x	2 ^{-17.9552}
508A71FF _x	-2 ^{-19.2861}	B5DAA46B _x	2 ^{-17.2169}	0DC239A7 _x	2 ^{-20.5299}
F8ED0A0B _x	2 ^{-18.7207}	411ED3B3 _x	2 ^{-21.0006}	94E844C3 _x	-2 ^{-20.1235}
F24337B1 _x	-2 ^{-18.8501}	BBE96696 _x	-2 ^{-18.1351}	F6148C2A _x	-2 ^{-18.3755}
0C34BDF3 _x	-2 ^{-17.4189}	32A5DEAF _x	-2 ^{-16.8141}	5EB3E9BB _x	2 ^{-18.3688}
398233B0 _x	2 ^{-17.6361}	FFD6F2AE _x	2 ^{-17.2869}	396A0040 _x	2 ^{-23.1944}
83FB82ED _x	2 ^{-18.6361}	E97815C2 _x	-2 ^{-19.7837}	24EC1390 _x	2 ^{-20.9910}
Run 7		Run 8		Run 9	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
A096F3E6 _x	-2 ^{-16.0597}	560533AC _x	-2 ^{-15.9254}	0841D2D3 _x	-2 ^{-15.5616}
D0A9D240 _x	-2 ^{-18.3818}	F8E3E4FB _x	-2 ^{-19.8774}	2A6A80BB _x	-2 ^{-19.0762}
46373DFE _x	2 ^{-18.1380}	DC190EF0 _x	-2 ^{-17.8428}	72A6FCD6 _x	2 ^{-20.3500}
15895B21 _x	-2 ^{-19.3119}	A5155394 _x	2 ^{-16.1876}	CAB51EF7 _x	2 ^{-17.2106}
5A403D62 _x	-2 ^{-18.5704}	01061496 _x	2 ^{-16.8843}	D1B4AE5D _x	2 ^{-17.8585}
2E66E414 _x	2 ^{-21.1715}	3FB87759 _x	-2 ^{-17.1370}	21284C10 _x	2 ^{-18.0342}
04DD66C5 _x	2 ^{-21.0084}	E08EBDF0 _x	2 ^{-19.8748}	D9BD6A99 _x	2 ^{-19.2286}
2BB84DFD _x	2 ^{-19.7062}	2DDF9477 _x	2 ^{-18.2082}	131D8ABD _x	2 ^{-18.4165}
BAE4A96E _x	-2 ^{-17.4475}	FDC770D4 _x	2 ^{-21.4741}	040D421A _x	-2 ^{-17.3171}
5507078F _x	2 ^{-18.9505}	0B38CBE3 _x	-2 ^{-18.8104}	1F7590A3 _x	2 ^{-17.0995}
B299AD82 _x	2 ^{-17.7568}	C703D7F9 _x	2 ^{-20.0152}	0BC17C43 _x	-2 ^{-17.0447}
6B639FC2 _x	-2 ^{-18.9349}	01B81B61 _x	2 ^{-21.1555}	94A8D2D6 _x	2 ^{-18.6436}
1E04D922 _x	-2 ^{-18.7995}	C3A5180C _x	2 ^{-19.2615}	3F9CA5E4 _x	2 ^{-17.6383}
2CB164AE _x	-2 ^{-19.9789}	AB74FC57 _x	2 ^{-17.1102}	A64226DE _x	2 ^{-18.3055}
782AC4B9 _x	2 ^{-19.8064}	0969D431 _x	2 ^{-20.6310}	814D89A2 _x	-2 ^{-19.4767}
B2C73118 _x	-2 ^{-17.9784}	702CB326 _x	-2 ^{-19.3912}	40C22B2A _x	2 ^{-21.6249}
Run 10		Run 11		Run 12	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
B9E85A1A _x	2 ^{-17.0648}	2EB55D68 _x	2 ^{-15.8385}	4F3CEE16 _x	2 ^{-15.8386}
DD399B66 _x	2 ^{-18.2499}	8DA98656 _x	2 ^{-21.5538}	6118C102 _x	2 ^{-20.1948}
C4712160 _x	2 ^{-17.6473}	7F5C9161 _x	2 ^{-20.5999}	5EDE09FA _x	2 ^{-16.4085}
0E845DE9 _x	-2 ^{-17.8888}	E2E3F48B _x	-2 ^{-19.3037}	C6D23AF3 _x	-2 ^{-19.3290}
F5697A15 _x	2 ^{-16.6692}	3D9CD58A _x	2 ^{-21.4999}	917BDE76 _x	2 ^{-19.2828}
20AE2D1E _x	-2 ^{-18.6498}	E3232ED1 _x	-2 ^{-17.1821}	DB9D1706 _x	-2 ^{-16.6047}
4E1A9439 _x	-2 ^{-20.4738}	D215324F _x	2 ^{-21.3974}	B52017DF _x	-2 ^{-17.9765}
A692CF68 _x	-2 ^{-25.7297}	68CB1CBB _x	2 ^{-21.0152}	E932226C _x	-2 ^{-17.3126}
2D6E74B9 _x	-2 ^{-19.1425}	6E3AF545 _x	2 ^{-17.3588}	798D01C6 _x	2 ^{-17.5631}
4BA432CE _x	2 ^{-24.1769}	E06121A4 _x	-2 ^{-19.7149}	8C374DF8 _x	2 ^{-17.3292}
908FB0E1 _x	2 ^{-16.1571}	F5DA598A _x	-2 ^{-17.7904}	CC6C8D7F _x	-2 ^{-17.0031}
87D2C47D _x	-2 ^{-17.7923}	9E0718E9 _x	2 ^{-23.8105}	431CAB81 _x	2 ^{-17.5546}
45BBB748 _x	2 ^{-17.2851}	FE5FB280 _x	-2 ^{-18.7501}	5D4A703F _x	-2 ^{-18.0655}
EF778233 _x	-2 ^{-17.9270}	8650F29D _x	-2 ^{-19.6677}	C0C4F6E0 _x	-2 ^{-18.2996}
AB448DE7 _x	2 ^{-19.0715}	52EEFA5D _x	2 ^{-18.6606}	FF5CC8D0 _x	2 ^{-19.7086}
B35C0B5E _x	-2 ^{-18.4746}	DDBBCC6 _x	2 ^{-19.1039}	8AEE31A7 _x	2 ^{-21.3384}

Table 14: Results Experiment 1, run 13-14

Run 13		Run 14	
Class	Cond. corr.	Class	Cond. corr.
05731E8B _x	$-2^{-16.2620}$	D124C4C7 _x	$2^{-15.9330}$
5DB0E778 _x	$-2^{-17.5226}$	1EAB8149 _x	$2^{-18.2779}$
1978A847 _x	$2^{-16.4214}$	D096E50A _x	$-2^{-16.8167}$
0714A162 _x	$2^{-17.4079}$	F63D83A0 _x	$2^{-16.7964}$
E427DDA1 _x	$-2^{-19.0854}$	58DD999F _x	$-2^{-19.6575}$
F1043FF0 _x	$2^{-18.4435}$	687BD459 _x	$2^{-17.2279}$
2063BF16 _x	$2^{-18.2604}$	7ED57159 _x	$-2^{-18.1704}$
72FEEE44 _x	$2^{-16.8421}$	3E16ED13 _x	$-2^{-19.2129}$
84FB3EB8 _x	$2^{-18.1412}$	AB7C3CDF _x	$2^{-20.4807}$
D460EBE6 _x	$2^{-19.2130}$	5B44777F _x	$2^{-17.3261}$
8B520DDC _x	$-2^{-18.5055}$	EB21F368 _x	$2^{-17.1960}$
62AD0B93 _x	$2^{-17.6416}$	8D37614E _x	$-2^{-18.7395}$
49724508 _x	$2^{-18.2654}$	D8D92EA8 _x	$2^{-18.2482}$
961235FB _x	$2^{-18.9537}$	53E8CFA4 _x	$2^{-21.3423}$
1B2DF449 _x	$-2^{-16.7523}$	4BFDC6C2 _x	$2^{-19.8957}$
9439837B _x	$2^{-17.2473}$	F0873852 _x	$-2^{-17.5521}$

Table 15: Results Experiment 1, run 15-16

Run 15		Run 16	
Class	Cond. corr.	Class	Cond. corr.
CDE8ACA5 _x	$-2^{-16.5740}$	CA1D403C _x	$-2^{-15.6369}$
CAED06DE _x	$-2^{-17.8129}$	269AEC23 _x	$2^{-17.6759}$
54B90F29 _x	$2^{-18.5584}$	18A8ABD9 _x	$2^{-20.1006}$
4CA9E3D1 _x	$2^{-18.3883}$	F6624A07 _x	$2^{-17.8805}$
6CDE2F4A _x	$2^{-19.2274}$	39526EEB _x	$-2^{-18.4606}$
EA60B0CC _x	$-2^{-19.0488}$	86EC1AFE _x	$2^{-16.7075}$
71848C19 _x	$2^{-19.7642}$	E7D25F0A _x	$-2^{-17.4129}$
4B2F81CD _x	$-2^{-18.2663}$	79C1D69B _x	$2^{-18.2451}$
F202E2B0 _x	$2^{-17.9375}$	794E2396 _x	$-2^{-18.2178}$
A799466F _x	$-2^{-17.3837}$	08966969 _x	$2^{-17.8062}$
2EF92DBD _x	$-2^{-20.7477}$	D6D50228 _x	$-2^{-19.1594}$
177B5383 _x	$2^{-16.9851}$	4FD5EEF1 _x	$2^{-17.6230}$
3ED126D7 _x	$2^{-16.7947}$	F2D8D95A _x	$2^{-17.8633}$
6F7C522C _x	$2^{-17.9543}$	8DE104EC _x	$-2^{-20.9480}$
3B690371 _x	$2^{-19.4156}$	AE59DF9B _x	$2^{-16.6174}$
398B8EAE _x	$-2^{-17.8491}$	9CDF42F0 _x	$-2^{-18.5145}$

Table 16: Settings Experiment 2

run	key k																substantial class $\bar{X}(k_0)$
1	D2	E7	BD	7E	06	4A	6F	15	AE	5D	39	B9	56	FA	50	C1 _x	7739 _x
2	AB	78	62	7C	36	16	BB	8B	F3	7D	55	BB	00	D0	F7	9A _x	E1DF _x
3	CA	43	A2	7A	B5	4B	67	77	D5	65	E3	4C	98	FD	0A	01 _x	1E03 _x
4	5C	FF	CC	D5	A7	C6	B5	DF	3D	08	74	66	4B	B5	5F	05 _x	242F _x
5	87	25	84	F0	51	5B	BD	45	82	75	B1	F7	37	98	40	31 _x	F0E3 _x
6	FD	AC	D1	33	A2	EB	0D	C5	07	48	00	33	51	C2	61	0E _x	E2B1 _x
7	E2	79	A6	01	48	F7	74	F9	78	8C	E9	02	D5	A6	25	DA _x	C40E _x
8	62	AC	EC	54	68	92	18	B4	F7	DD	92	8E	D1	66	D4	2D _x	7A7B _x
9	06	93	63	2A	8A	34	1A	E9	E9	00	24	3B	4C	77	CD	BE _x	A4F4 _x
10	10	49	22	10	34	A5	68	63	2E	A5	BE	17	96	90	C5	76 _x	72A9 _x
11	40	C3	EE	A1	05	55	F5	C5	1C	78	C8	69	35	32	61	BB _x	9A05 _x
12	AA	95	B1	8D	AF	97	00	36	84	2B	BF	8D	3F	F4	6C	D2 _x	5185 _x
13	82	3B	85	7F	4C	12	9A	8A	E1	12	8D	1C	2C	C9	1D	45 _x	AD96 _x
14	81	05	9B	F3	F8	7F	8F	8C	87	D3	4E	97	0B	29	DB	97 _x	1411 _x
15	06	0A	9E	84	FF	39	1B	EF	5B	4E	05	FC	32	8F	EE	A4 _x	CBF4 _x
16	32	91	C5	32	60	EF	D4	B3	9C	FA	DD	A7	1C	4D	85	04 _x	8696 _x

Table 17: Results Experiment 2, run 1-4

Run 1		Run 2		Run 3		Run 4	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
7739 _x	2 ^{-14.7755}	E1DF _x	2 ^{-14.4051}	1E03 _x	-2 ^{-14.2988}	242F _x	2 ^{-14.3707}
0BA7 _x	-2 ^{-15.1016}	DECD _x	2 ^{-14.7583}	BA93 _x	2 ^{-15.4010}	388E _x	2 ^{-14.6176}
C56D _x	-2 ^{-15.9418}	8BD2 _x	2 ^{-15.9473}	95E2 _x	-2 ^{-15.5154}	C034 _x	2 ^{-16.1970}
E0A4 _x	-2 ^{-17.5555}	8F18 _x	-2 ^{-16.6231}	47F4 _x	-2 ^{-16.1295}	632E _x	-2 ^{-17.6330}
B205 _x	-2 ^{-16.7841}	7CDF _x	-2 ^{-15.9235}	5722 _x	2 ^{-17.3693}	E279 _x	2 ^{-18.7117}
DBCF _x	-2 ^{-14.9386}	911B _x	2 ^{-18.5230}	322D _x	2 ^{-15.8003}	B28C _x	2 ^{-15.4069}
94A2 _x	-2 ^{-15.9209}	F4DB _x	-2 ^{-15.4753}	EF5B _x	-2 ^{-17.0102}	26C4 _x	-2 ^{-18.8490}
E406 _x	2 ^{-16.3749}	FE17 _x	-2 ^{-16.0871}	D8AD _x	2 ^{-18.5977}	CB00 _x	-2 ^{-16.4241}
C716 _x	-2 ^{-15.4192}	3B8A _x	-2 ^{-19.1257}	8555 _x	-2 ^{-18.7710}	2CCF _x	-2 ^{-16.1990}
3D7C _x	-2 ^{-16.1821}	6731 _x	2 ^{-17.3923}	9EFB _x	-2 ^{-20.1980}	86F8 _x	2 ^{-14.8792}
DFC8 _x	-2 ^{-20.0540}	AEF9 _x	-2 ^{-16.2686}	E725 _x	2 ^{-15.8796}	4E16 _x	-2 ^{-19.1657}
CE69 _x	-2 ^{-15.0659}	264F _x	-2 ^{-17.1391}	A5DB _x	-2 ^{-15.4151}	F68E _x	-2 ^{-16.1821}
E317 _x	-2 ^{-19.4358}	D568 _x	2 ^{-16.3593}	6F63 _x	-2 ^{-15.6374}	DC87 _x	-2 ^{-15.2997}
ABB9 _x	2 ^{-18.2686}	1C8C _x	2 ^{-16.2737}	C883 _x	-2 ^{-16.7736}	4D0B _x	2 ^{-14.3464}
EA58 _x	-2 ^{-16.5159}	57A9 _x	2 ^{-16.0766}	BC3C _x	2 ^{-16.6672}	F52F _x	-2 ^{-16.4191}
68A3 _x	2 ^{-18.4225}	6BF4 _x	2 ^{-16.5487}	8F28 _x	2 ^{-18.9721}	CA7D _x	-2 ^{-17.5357}

Table 18: Results Experiment 2, run 5-16

Run 5		Run 6		Run 7		Run 8	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
F0E3 _x	2 ^{-14.0013}	E2B1 _x	2 ^{-14.3716}	C40E _x	2 ^{-14.4449}	7A7B _x	2 ^{-14.8248}
DB7E _x	-2 ^{-15.7332}	8CA5 _x	2 ^{-20.0693}	E7A6 _x	2 ^{-16.8690}	42BE _x	2 ^{-17.1346}
DB49 _x	-2 ^{-16.4554}	5722 _x	2 ^{-15.0597}	C134 _x	-2 ^{-15.3590}	A34A _x	2 ^{-18.5138}
BCE7 _x	-2 ^{-17.4877}	B362 _x	2 ^{-14.9223}	935D _x	2 ^{-15.4809}	C238 _x	2 ^{-15.0471}
B219 _x	2 ^{-20.7283}	A6FF _x	2 ^{-16.5127}	7478 _x	-2 ^{-16.4987}	EDE1 _x	2 ^{-21.9871}
35EC _x	2 ^{-14.6546}	17A2 _x	-2 ^{-15.6205}	190B _x	-2 ^{-15.0138}	F6B0 _x	2 ^{-16.1473}
3A19 _x	2 ^{-16.6525}	80E6 _x	2 ^{-18.5839}	7DC3 _x	2 ^{-16.3652}	0BE1 _x	2 ^{-16.8602}
ECC4 _x	-2 ^{-15.5898}	0C8E _x	2 ^{-14.6881}	4B0B _x	2 ^{-16.8354}	1F33 _x	2 ^{-16.0671}
C1E4 _x	2 ^{-15.0888}	9689 _x	-2 ^{-16.8477}	B242 _x	-2 ^{-16.6587}	6EE5 _x	2 ^{-17.2484}
2043 _x	-2 ^{-16.3024}	87E0 _x	-2 ^{-15.1826}	1AA0 _x	2 ^{-15.8424}	A1F7 _x	2 ^{-16.3617}
F1C9 _x	2 ^{-16.2757}	A6C3 _x	2 ^{-15.4206}	214B _x	-2 ^{-18.2526}	0B3E _x	2 ^{-15.7595}
6CBC _x	-2 ^{-16.2378}	3385 _x	2 ^{-18.8934}	4317 _x	-2 ^{-15.8732}	A420 _x	-2 ^{-15.5446}
2815 _x	-2 ^{-16.2954}	CD7F _x	-2 ^{-18.9379}	1D46 _x	-2 ^{-16.8654}	EA4B _x	2 ^{-14.9606}
475F _x	-2 ^{-16.1167}	6E2E _x	2 ^{-16.8961}	2CFC _x	-2 ^{-16.9402}	3880 _x	2 ^{-15.8658}
697A _x	-2 ^{-15.2881}	E2DA _x	2 ^{-17.4384}	24F4 _x	2 ^{-15.9180}	16B1 _x	2 ^{-19.0984}
988D _x	-2 ^{-16.9645}	530C _x	2 ^{-15.6815}	BC3B _x	2 ^{-15.1942}	FC8A _x	-2 ^{-16.5374}
Run 9		Run 10		Run 11		Run 12	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
A4F4 _x	-2 ^{-14.4412}	72A9 _x	-2 ^{-15.0900}	9A05 _x	2 ^{-13.9470}	5185 _x	2 ^{-13.9585}
19A8 _x	-2 ^{-17.2236}	6FEC _x	2 ^{-18.3103}	BAFE _x	-2 ^{-15.2182}	F7EC _x	-2 ^{-15.2780}
3B6D _x	-2 ^{-14.5100}	C936 _x	2 ^{-16.7246}	0203 _x	-2 ^{-15.5227}	B9AE _x	2 ^{-18.4358}
AC88 _x	-2 ^{-16.3749}	3598 _x	-2 ^{-15.5514}	3ED5 _x	2 ^{-15.2702}	9AAA _x	2 ^{-17.0183}
37D2 _x	-2 ^{-17.7640}	CF43 _x	-2 ^{-16.0569}	CA16 _x	-2 ^{-18.3100}	C141 _x	2 ^{-14.5610}
1150 _x	2 ^{-15.6985}	868B _x	-2 ^{-15.8424}	8524 _x	2 ^{-16.1654}	7C9D _x	-2 ^{-15.5436}
5F5F _x	2 ^{-16.0560}	0FA7 _x	-2 ^{-14.9989}	68AB _x	2 ^{-18.6388}	2161 _x	2 ^{-18.0580}
BBCB _x	-2 ^{-15.0803}	2F07 _x	-2 ^{-22.2192}	435F _x	-2 ^{-18.9816}	48E9 _x	2 ^{-15.6957}
BC5C _x	-2 ^{-15.2454}	45B6 _x	-2 ^{-16.5485}	4A7F _x	2 ^{-17.6849}	404B _x	-2 ^{-15.6813}
B564 _x	2 ^{-14.9222}	243A _x	2 ^{-15.2239}	47BF _x	2 ^{-14.7445}	42AF _x	2 ^{-15.5912}
1293 _x	-2 ^{-16.1343}	DB15 _x	-2 ^{-15.8345}	3951 _x	2 ^{-17.2349}	BE9D _x	2 ^{-17.0223}
21D1 _x	-2 ^{-18.2133}	51F4 _x	-2 ^{-14.8893}	371A _x	-2 ^{-18.2964}	174A _x	-2 ^{-15.6898}
5D60 _x	-2 ^{-16.2374}	1358 _x	-2 ^{-15.6970}	C3B9 _x	2 ^{-17.2509}	E7C5 _x	-2 ^{-18.7995}
F344 _x	-2 ^{-16.5596}	2315 _x	2 ^{-15.9978}	DDE4 _x	-2 ^{-18.0504}	2A2C _x	-2 ^{-15.9797}
9CF2 _x	-2 ^{-15.5289}	8A7F _x	-2 ^{-17.4412}	6E09 _x	2 ^{-18.7562}	9C3D _x	-2 ^{-15.5690}
BD63 _x	2 ^{-14.2884}	44B1 _x	-2 ^{-15.8216}	4ABD _x	2 ^{-18.1714}	2FC7 _x	-2 ^{-16.4663}
Run 13		Run 14		Run 15		Run 16	
Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.	Class	Cond. corr.
AD96 _x	2 ^{-14.1913}	1411 _x	-2 ^{-13.4589}	CBF4 _x	2 ^{-13.8074}	8696 _x	2 ^{-13.7917}
FF04 _x	-2 ^{-18.6912}	775C _x	2 ^{-15.7345}	7DB0 _x	-2 ^{-17.2316}	F63F _x	-2 ^{-15.7935}
908D _x	-2 ^{-15.3723}	01A3 _x	-2 ^{-17.8515}	3B6A _x	2 ^{-16.4807}	63E2 _x	2 ^{-14.6600}
BD3C _x	2 ^{-19.0892}	65E1 _x	2 ^{-15.1726}	7B79 _x	2 ^{-16.0247}	404F _x	-2 ^{-16.3921}
2684 _x	-2 ^{-14.8869}	FB61 _x	2 ^{-14.9914}	9D17 _x	2 ^{-19.4792}	62E4 _x	2 ^{-15.5395}
4622 _x	-2 ^{-17.4602}	C750 _x	-2 ^{-17.0675}	3CB1 _x	-2 ^{-16.5038}	97D7 _x	2 ^{-17.1093}
B95E _x	-2 ^{-18.4386}	4714 _x	2 ^{-14.8149}	8570 _x	-2 ^{-15.8811}	00C0 _x	-2 ^{-17.1480}
3409 _x	-2 ^{-17.0849}	CA10 _x	-2 ^{-19.6321}	FC18 _x	-2 ^{-15.0194}	A35A _x	-2 ^{-18.0227}
A1E8 _x	-2 ^{-16.8910}	C2B3 _x	-2 ^{-17.9569}	C44B _x	-2 ^{-16.7061}	61E0 _x	-2 ^{-20.2835}
B4AB _x	-2 ^{-17.1547}	E9AE _x	2 ^{-16.3241}	CE37 _x	2 ^{-16.1367}	1D38 _x	-2 ^{-16.7416}
C844 _x	2 ^{-15.8789}	5B12 _x	-2 ^{-18.0958}	EADE _x	-2 ^{-16.8083}	7C57 _x	-2 ^{-15.6409}
417A _x	2 ^{-18.2304}	BB3B _x	-2 ^{-14.4584}	AFC3 _x	-2 ^{-17.6059}	1A3D _x	2 ^{-21.1668}
152B _x	2 ^{-16.5412}	04A8 _x	2 ^{-18.2655}	CC21 _x	-2 ^{-14.7704}	3548 _x	2 ^{-16.1807}
F68F _x	-2 ^{-16.4105}	684F _x	-2 ^{-16.3141}	3EA8 _x	-2 ^{-17.2445}	8A66 _x	-2 ^{-19.0099}
D00F _x	-2 ^{-18.5125}	9648 _x	2 ^{-17.7090}	1A4D _x	-2 ^{-22.6600}	F063 _x	-2 ^{-17.0442}
49A0 _x	-2 ^{-16.0733}	F1D2 _x	-2 ^{-16.4496}	C6CD _x	-2 ^{-16.1096}	BB46 _x	-2 ^{-16.0631}